

TCG EK Credential Profile

For TPM Family 2.0; Level 0

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RC1

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Work in Progress

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Public Review

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Change History

Revision	Section	Description
V2.0 R14 (Nov 2014)		First public version of this specification for TPM 2.0
V2.1 R13 (Dec 2018)	2.2.1 Annex B, C Annex B.4 3.2	Incorporated Errata for handling of NV Index Handles Moved algorithm-specific content to annexes Added new EK Templates in the High Range for RSA 2048, ECC NIST P256, NIST P384, NIST P521, and SM2_256 Relaxed EK Certificate fields: - TPM spec version changed from MUST to MAY - TPM security assertions changed from MAY to SHOULD NOT - HardwareModuleName changed from MAY to SHOULD NOT - Certificate Policies changed from SHOULD to MAY, and policy qualifiers from MAY to SHOULD NOT - Authority Information Access (AIA) changed from SHOULD to MAY - Extended Key Usage changed from SHOULD to MAY
V2.3 R2 (July 2020)	2.2.1.5.2 2.2.1.5.1 2.2.1.5.1, B.4.4.6 & 7, C.1.1.2	Added NV handles for EK Certificate chain intermediates Changed handle values for EK Certificates in the High Range to be normative Added support for RSA 3k and 4k: - NV Index handles for EK Certificates - EK Templates and - signature algorithms
V2.4 R3 (July 2021)	2.2.1.4, 2.2.1.5 3.2.13 3.2.16	Removed all “For TPMs designed to meet Windows” statements, which are: - RSA 2k and ECC NIST P256 EK must use low range EK Templates, and associated EK Certificates must be stored using low range handles - EK Certificate must contain AIA extension that contains URL for the issuing CA Certificate - EK Certificate must contain ECU extension that contains OID tcg-kp-EKCertificate Reader should consult respective operating system requirements instead
V2.5 R2	2.2.1.5, 2.2.1.9	Exclude range of NV indices used for certificate chains from valid range for EK Certificates.
V2.6	2.2.1 3.1.1 2.2.2.5.1 B.3, B.4.4 B.4	Added requirements for firmware-limited EK credentials Added <i>firmwareLimited</i> to ekGenerationType Added NV Index handle values for firmware-limited and Signing EK Certificates Updated tables for all existing templates (L-1 to L-2, H-1 to H-7) to remove absent fields (with NULL assigned as content), and added previously reserved object attributes (<i>firmwareLimited</i> and <i>svnLimited</i>); Templates are not changed Added Templates for firmware-limited and Signing EKs
V2.7	2.2.2.5.1	Added NV Index handle values for ML-KEM and ML-DSA EK certificates

	3.1.4 & 5, 3.2.11.2 & 3	Added EK Credential Algorithm List and TPM PQC Version attributes
	3.2.15	Added that the keyEncipherment bit must also be set for an ML-KEM EK certificate
	5.4.6.5, 5.4.6.6	Added Templates for ML-KEM (Storage and firmware-limited Storage) and ML-DSA (Signing and firmware-limited Signing)
	6.1.3	Added definition for ML-DSA signature algorithm
	6.2.3, 6.2.4	Added definition for ML-KEM and ML-DSA subject public key info

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1 Introduction

1.1 Purpose

The purpose of this document is to define the TPM 2.0 Endorsement Key (EK) Credential. This specification describes the content of the credential and provides an X.509 instantiation of the credential. A standardized and commonly used format should provide better interoperability between credential providers and users.

1.2 Scope

This document specifies the TPM 2.0 Endorsement Key Credential. It does not apply to TPM 1.2 credentials or credentials of other type.

1.3 Relationship to Other TCG Specifications

A TPM claiming adherence to this specification SHALL be compliant with the TPM 2.0 Library Specification[1]; Family 2.0; Level 00; Revision 00.99 or later.

1.4 Keywords

The key words “MUST”, “MUST NOT”, “REQUIRED”, “SHALL”, “SHALL NOT”, “SHOULD”, “SHOULD NOT”, “RECOMMENDED”, “MAY”, and “OPTIONAL” in this document are to be interpreted as described in RFC 2119[17].

1.5 Statement Type

Please note a very important distinction between different sections of text throughout this document. There are two distinctive kinds of text: informative comment and normative statements. Because most of the text in this specification will be of the kind normative statements, the authors have informally defined it as the default and, as such, have specifically called out text of the kind informative comment. They have done this by flagging the beginning and end of each informative comment and highlighting its text in gray. This means that unless text is specifically marked as of the kind informative comment, it can be considered a kind of normative statement.

Start of informative comment

This is the first paragraph of 1–n paragraphs containing text of the kind *informative comment* ...

This is the second paragraph of text of the kind *informative comment* ...

This is the nth paragraph of text of the kind *informative comment* ...

To understand the TCG specification the user must read the specification. (This use of MUST does not require any action).

End of informative comment

An additional informative statement type is presented as follows:

“Note: ...”

This informative statement type provides additional context in line with and immediately following a normative requirement.

1.6 Abbreviations

CFB Cipher Feedback mode

CSR	Certificate Signing Request
DER	Distinguished Encoding Rules
DICE	Device Identifier Composition Engine
EK	Endorsement Key
EPS	Endorsement Primary Seed
IDevID	Initial Device Identifier
KDF	Key Derivation Function
OCSP	Online Certificate Status Protocol
OEM	Original Equipment Manufacturer
RDN	Relative Distinguished Name
TPM2_	Prefix that indicates a TPM 2.0 command

1.7 Definition of Terms

The TCG Glossary [20] contains a few definitions that are fundamental to this document.

The following operational definitions, however, are specific to this specification.

Certificate – A certificate is an instantiation of a credential using the industry-standard certificate structure from ISO/IEC/ITU-T X.509 version 3 [27]. Certificate generation consists of (a) assembling values for the credential fields and (b) signing over the assembled fields.

Credential – A credential is an abstract proof that must be instantiated as a certificate before it can be exchanged between entities.

2 TPM 2.0 EK and EK Credential

2.1 Endorsement Key

Start of informative comment

The Endorsement Key (EK) is an asymmetric key pair consisting of a public and private key stored in a Shielded Location on the TPM. The public part of the EK can be read from the TPM while the private part is never exposed. The public key of the EK is included in the EK Certificate.

In TPM 1.2, the Endorsement Key was defined as an RSA 2048 bit key. This is not the case for TPM 2.0, which can have more than one EK. The algorithm flexibility provided by the TPM 2.0 Library Specification [1] allows the TPM to create EKs of any type of asymmetric algorithm implemented in the TPM (see 2.1.1). The properties of the Endorsement Key are defined by its public area structure, the “template”. This specification defines multiple default templates for algorithm-specific Endorsement Keys.

End of informative comment

Any asymmetric algorithm supported in a platform-specific specification used to implement the TPM MAY be used as a key type instead of or in addition to the defined key types in this document.

A relevant Platform specification can require that an EK be persisted in the TPM when it ships.

2.1.1 Primary Key Generation

Start of informative comment

The Endorsement Key is a Primary Object controlled by the Endorsement Hierarchy. The Endorsement Hierarchy has a Seed, the Endorsement Primary Seed (EPS), which is unique to each TPM. The Primary Seed is a large random value; its size is required to be at least twice the security strength of any algorithm implemented on the TPM. The TPM Library Specification requires the EPS be generated within the TPM or be generated and injected by the TPM manufacturer in the manufacturing environment. The attribute `TPMA_PERMANENT.tpmGeneratedEPS` (see *TPM 2.0 Library Specification, Part 2 [1]*) is set to indicate the source of the Seed. The attribute can be read with `TPM2_GetCapability()`. The Seed cannot be read from the TPM and is never exposed.

The TPM 2.0 Library Specification [1] defines a process to create primary keys based on a Key Derivation Function (KDF) and a Primary Seed. The KDF is a deterministic function that uses key parameters to derive a reproducible key. These parameters determine the type of the key and are input to the command `TPM2_CreatePrimary()`. When this command is called with the same parameters, the same key is generated as long as the EPS does not change. The key can be made persistent in TPM NV memory using the command `TPM2_EvictControl()` or recreated when needed. This way, any type of key (e.g. symmetric or asymmetric, signing or decryption key) can be created by the TPM.

As of revision 1.38 of the TPM 2.0 Library Specification [1], `TPM2_CreatePrimary()` uses the sensitive data (provided in the *inSensitive.sensitive.data* parameter) as part of the calculation for the Primary Key.

End of informative comment

In the creation of an Endorsement Key for which an EK Credential is issued, the sensitive data size (*inSensitive.sensitive.data.t.size*) MUST be set to zero.

2.1.2 EK Usage

Start of informative comment

In TPM 1.2, the Endorsement Key was defined as a decryption key; it could not be used for signing operations. Unlike TPM 1.2, TPM 2.0 provides more flexibility in defining an EK. The properties of the Endorsement Key are determined by its public area template (`TPMT_PUBLIC` structure, see *TPM*

2.0 Library Specification Part 2 [1]). The TPMT_PUBLIC structure includes the base attributes restricted, sign and decrypt that determine the cryptographic operation a key may perform on an object. The TPM 2.0 Library Specification [1] does not impose any restrictions regarding the attributes of the Endorsement Key. As any other key the Endorsement Key can be created as a decryption or signing key.

However, the EK and its credential may be considered privacy-sensitive if the private part of the EK is used in a cryptographic protocol. In this case, the public EK or the EK Certificate could represent a privacy-sensitive cryptographic identifier for a particular platform. In privacy-sensitive environments, the EK should not be a signing key and restricted to specific operations (this is described in more detail in section 2.3 Privacy Protection).

On the other hand, there are environments where privacy is not an issue. This specification distinguishes between user device TPMs and non-user device TPMs. Whether the EK may sign depends on the type of platform for which the TPM is built.

End of informative comment

2.1.2.1 User Device TPM

Start of informative comment

User device TPMs are TPMs that can be associated with a human user, typically in a PC Client or Mobile platform. In this case, the EK is best defined as a restricted decryption (Storage) key.

End of informative comment

2.1.2.2 Non-User Device TPM

Start of informative comment

Non-user device TPMs on the other hand are TPMs that are associated with an enterprise, rather than a specific user. This can be, e.g.,

- Network Elements (e.g. routers, switches, wireless access points)
- Servers, Virtual Servers, Virtual Devices in a cloud infrastructure
- Embedded Devices (e.g. printers)

For such platforms, privacy is not a central concern and unique identification is of critical importance. These platforms can use a certified EK for signing operations. This is intended to facilitate the establishment of further TPM keys, like Device Identification keys, without the need for an Attestation Key. This allows for simpler infrastructure implementations. In this case, no restrictions other than those defined in the TPM 2.0 Library Specification [1] apply to the settings of the base attributes. Legacy EKs can be defined as general-purpose keys if both signing and decryption should be supported. This is not true of Post-Quantum EKs. The Post-Quantum Cryptographic (PQC) algorithms are not implemented to allow both decryption and signing in the same algorithm. The key usage field in the EK Credential defined in section 3.2.15 is set appropriately to indicate the usage of the EK.

One use case for a signing EK is to sign the Certificate Signing Request (CSR) for an Initial Device Identifier (IDevID) key. The IDevID key is a TPM-generated key that is used as an initial identity for secure device authentication (see IEEE 802.1AR [10]). The CSR can be signed with the command TPM2_Sign(). The hash calculated over the certification request information is passed to the TPM in the digest command parameter; the *inScheme* command parameter specifies the signing scheme. The issuer of the IDevID Certificate could verify the CSR signature to ensure that the chip requesting the IDevID Certificate is privileged to receive it. Therefore, the issuer could have a list of EK Certificates of all valid TPMs a product manufacturer has purchased. Alternatively, a CSR could be signed by an Attestation Key.

End of informative comment

2.1.3 EK Lifetime

Start of informative comment

In TPM 2.0, the lifetime of an Endorsement Key is tied to the Endorsement Primary Seed (EPS). As long as the EPS is not changed, EKs can be recreated with their public area templates. The command TPM2_ChangeEPS() replaces the Endorsement Primary Seed with a new random value and makes it impossible to recreate any EKs derived from the previous Seed. This will invalidate all certificates associated with the EKs and will affect Platform Certificates that reference EK Certificates.

Platform-specific specifications determine whether the command TPM2_ChangeEPS() is required to be implemented. Some platforms might want to change the EPS, e.g. during platform refurbishment to erase existing EKs or after a field upgrade from a firmware that had a severe security flaw (in order to revoke all EKs associated with the old firmware).

On the other hand, there are platforms that need a permanent EPS because invalidating the Endorsement Keys would prevent the platform proving that it is a genuine trusted platform. In non-user device TPMs (see 2.1.2.2), for instance, the EPS is required to be permanent because the EK represents the trust anchor for the device identity.

The TPM 2.0 Library Specification [1] provides a means to prevent the EKs from being replaced. The command to change the EPS requires Platform Authorization, so the OEM can decide if the EPS ever changes. The use of Platform Authorization can be disallowed by turning off the Platform hierarchy (by setting the *phEnable* flag to CLEAR); this disables any functionality in the TPM that would require *platformAuth* or *platformPolicy*. Furthermore, TPM2_ChangeEPS() can be added to the list of commands that require assertion of Physical Presence using TPM2_PP_Commands(). Alternatively, platforms could prevent EKs from being erased by not exposing this functionality to the user.

Some TPMs may also support firmware-limited EKs. A TPM's firmware could be changed for a variety of reasons, such as improving existing features, adding new features, or correcting a critical security vulnerability. Firmware-limited EKs always change whenever the host TPM's firmware changes, irrespective of the reason for the change. Naturally, a change to a firmware-limited EK will always invalidate all extant certificates associated with that EK.

End of informative comment

2.2 Endorsement Key Credential

The Endorsement Key Credential is an X.509 v3 [12] certificate that contains the public EK, as well as various assertions regarding the security qualities and provenance of the TPM. The definitions of the certificate fields are specified in section 3.2. The EK Credential is usually issued by a TPM or Platform manufacturer during the manufacturing process. An entity SHALL NOT create an EK Credential for a TPM unless the entity is satisfied that the public key referenced in the EK Credential was either:

- returned in response to a TPM2_CreatePrimary() command by an implementation of Protected Capabilities and Shielded Locations that meets the TPM 2.0 Library Specification [1] or
- generated outside the TPM and inserted by a process defined in the Target of Evaluation (TOE) of the security target in use to evaluate the TPM.

There might be use cases where it is useful to issue an EK Credential after manufacturing (e.g. if the EPS was changed or the TPM is shipped without EK). In this case, the entity issuing the credential would create a new Endorsement Key with TPM2_CreatePrimary(). This procedure would require support for certificate enrollment. Support for an enrollment protocol is optional and can be done using a proprietary method of the TPM or Platform manufacturer or a method standardized by TCG. One example implementation (at time of writing, only available for TPM 1.2) is described in the IWG document *CMC Profile for EK/Platform Certificate Enrollment* [7].

A primary use case of an EK Credential is to assist Attestation CAs to issue credentials for restricted signing keys (Attestation Keys). The EK Credential can be used to provide evidence that the Attestation Keys are resident on the same TPM as the EK.

In TPM 2.0, multiple EKs can be derived from a single Seed (as described in section 2.1.1.) As a result, the TPM can have more than one EK Credential. However, the TPM might not be provisioned with all the credentials because of NV space restrictions; the credentials could be stored off the TPM. If an EK Credential is stored on the TPM, it is stored as an NV Index; in this case, it is referenced by its NV Index handle (see 2.2.2). The authorization to modify or access (read, write, delete) the credential is determined by the NV Index attributes. The attributes of the NV Index are defined by platform-specific workgroups, as well as the authorization for the index. Definitions specific to PC Client can be found in *PC Client Specific Platform TPM Profile for TPM 2.0* [6], section Non-volatile Storage.

2.2.1 Firmware-limited EKs

Start of informative comment

A TPM can support generation of firmware-limited EKs. Such keys have cryptographic values that depend upon the value of the TPM's firmware. As such, if the TPM firmware changes, a firmware-limited EK also changes. Firmware-limited EKs are a useful way of recovering trust in a TPM in exceptional circumstances where the TPM's firmware has a critical vulnerability that affects the trustworthiness of its EKs. See the TPM 2.0 Library Specification [1] for more details.

For example, to recover from a severe vulnerability, the TPM firmware can be upgraded and given a new firmware-limited EK. This EK, by virtue of being firmware-limited, cannot be accessed by the old, vulnerable TPM firmware. Firmware-limited EK credentials include measurements of the TPM firmware for which the EK was issued, and thus provide strong assurance as to the current version of TPM firmware.

This assurance relies on a credential being issued for the firmware-limited EK, which binds the EK public key to the firmware version that was used to derive the EK. A firmware limited EK addresses the threat model where the TPM firmware is not always well-behaved.

One approach is for the firmware-limited EK credential issuer to be an embedded CA running within the bootloader that is responsible for launching the TPM's firmware. In this case, the TPM bootloader is responsible for the following actions:

1. Cryptographically measuring the TPM's firmware before that firmware runs
2. Using that measurement, along with a secret known only to the TPM bootloader, to compute secret entropy that is revealed to TPM firmware
3. Using that secret entropy to derive a firmware-limited EK
4. Wielding a signing key known only to the bootloader to issue a credential for the EK, which certifies both the EK public key and the TPM firmware's measurements.

The TPM can cache firmware-limited EK credentials for use when the TPM firmware does not update between boots. If the TPM does not cache these credentials and instead generates them on each boot, a TPM's firmware-limited EK credential may change across boots even if the firmware does not change, if the signature algorithm is non-deterministic.

In order for users to trust the credential issued by the TPM bootloader, the bootloader's signing key will itself need to be endorsed by a trustworthy entity, such as the TPM vendor.

End of informative comment

The following list defines general requirements for the content of firmware-limited EK credentials.

1. Firmware-limited EK credentials SHALL include a measurement of the firmware that contributed to the derivation of the EK.
2. Firmware-limited EK credentials SHOULD include the TPM firmware's version and Security Version Number (SVN), as reported by the firmware itself.

The following list defines requirements for the content of firmware-limited EK credentials corresponding to TPM implementations based on Device Identifier Composition Engine (DICE) [23].

1. Firmware-limited EK credentials SHALL set *ekGenerationType* to *firmwareLimited* if TPM Security Assertions is included.
2. Firmware-limited EK credentials SHALL include a *tcg-dice-TcbInfo* extension or a *tcg-dice-MultiTcbInfo* extension [23].
3. The *version* and *svn* fields of the last evidence extension SHALL correspond with the TPM's self-reported version number and SVN.
4. The *fwids* field of the last evidence extension SHALL contain the cryptographic hash of the TPM firmware that was used to derive the secret entropy delivered to the TPM firmware.
5. DICE Name fields [23] are useful and SHOULD be set in the *TcbInfo*/*MultiTcbInfo* extension.
6. If the TPM does not report a firmware version and/or SVN, those respective fields SHALL NOT be populated in the DICE evidence extension.

Note: "last evidence extension" is defined as either:

- a. the *tcg-dice-TcbInfo* extension if the *tcg-dice-MultiTcbInfo* extension is not present, or
- b. the last *TcbInfo* embedded within the *tcg-dice-MultiTcbInfo* extension.

Note: an implementation may choose to embed a *MultiTcbInfo* extension, instead of a *TcbInfo* extension, in order to reflect a list of digests of firmware that contribute to the Trusted Computing Base of the TPM's application code. For more information see the "Multiple DiceTcbInfo Structures Extension" section of the DICE Attestation Architecture specification [23].

Note: an implementation may use a *TcbInfoAlias* extension instead of a *TcbInfo* extension. For more information see the "DiceTcbInfoAlias" section of the DICE Attestation Architecture specification [23].

To create a firmware-limited EK from a template, the user is required to create the EK in the *FW_ENDORSEMENT* hierarchy.

See TPM Part 1 [1] "Firmware-limited and SVN-limited Objects" for additional guidance on the use of firmware-limited EKs.

2.2.2 NV Index Handles

2.2.2.1 General Design

The NV Index handles related to the EK Credential MUST have values that are defined by the TCG in the *Registry of Reserved TPM 2.0 Handles and Localities* [2]. This section uses handles of the type "Global NV indices for OEMs, assigned by TCG", Refined Handle Type Endorsement Certificate.

This section defines two variations. The Low Range (see 2.2.2.4) generally includes the RSA 2048 and ECC NIST P256 EK data. The High Range (see 2.2.2.5) generally includes EK data for other algorithms. Both ranges permit the inclusion of an EK Template. In addition, the Low Range permits inclusion of a distinct EK Nonce. The High Range does not permit a distinct EK Nonce.

This document uses the following terms with specific meanings:

Term	Specific meaning within this document
<i>Absent</i>	This NV Index is not defined (i.e., this NV Index does not exist)
<i>Populated</i>	This NV Index is defined and written with content relevant to this document.

Within this document, manufacturer means either the TPM manufacturer or Platform manufacturer.

2.2.2.2 NV Index Contents

This section defines the NV Index contents and is applicable to both the Low Range and High Range.

Note: There is no length, type, or other metadata stored in the NV Index data.

2.2.2.2.1 EK Certificate

Start of informative

There is no assurance that the size of the NV Index is the same as the size of the encoded (using Distinguished Encoding Rules DER) certificate data. The Index may have additional bytes after the DER-encoded certificate data, and the value of those bytes is unspecified.

End of informative

An EK Certificate stored as an X.509 certificate in an NV Index SHALL be encoded in DER format.

2.2.2.2.2 EK Template

An EK Template is stored in an NV Index as a TPMT_PUBLIC structure marshaled as described in the TPM 2.0 Library Specification [1]. The default EK Templates are defined in section 5. The EK Template NV Index MUST be *Populated* if non-default values are used. It SHOULD be *Absent* if default values are used.

The EK Template unique field buffer size(s) SHOULD be zero.

Note 1: Setting the unique field buffer size(s) to zero minimizes the use of NV space. The unique field is only necessary to generate multiple different EKs if the rest of the TPMT_PUBLIC is the same. If other portions of the TPMT_PUBLIC change (e.g., the *authPolicy*), a different EK will be generated even if the unique field size(s) is zero.

Note 2: A platform-specific working group can define a proprietary default EK public area template that can be used instead of the templates defined in section 5.

Note 3: Users are advised that if they choose to create a Primary Key in the Endorsement Hierarchy that is different from the EK endorsed by the TPM manufacturer, they are strongly recommended to use a different template, and to utilize the *unique* field to provide cryptographic diversity.

2.2.2.2.3 EK Nonce

Start of informative

An EK Nonce is stored in an NV Index and is used to modify the EK Template *unique* field. The EK Nonce size is determined by the size field of the NV Index public area.

End of informative

2.2.2.3 Allowed and Recommended Usages of NV Indices

This table defines the allowed configurations of EK Nonce and EK Template in the Low and High ranges.

Table 1 EK Nonce and EK Template NV Indexes

EK Nonce	EK Template	Low Range Requirement	High Range Requirement
NV index <i>Absent</i>	NV index <i>Absent</i>	SHOULD	SHOULD
NV index <i>Absent</i>	NV index <i>Populated</i>	SHOULD NOT	See 2.2.2.5
NV index <i>Populated</i>	NV index <i>Absent</i>	MUST NOT	MUST NOT
NV index <i>Populated</i>	NV index <i>Populated</i>	SHOULD NOT	MUST NOT

2.2.2.4 Low Range

The Low Range SHALL use NV Indices 0x01c00002 - 0x01c0000c as defined below.

0x01c00002	RSA 2048 EK Certificate
0x01c00003	RSA 2048 EK Nonce
0x01c00004	RSA 2048 EK Template
0x01c0000a	ECC NIST P256 EK Certificate
0x01c0000b	ECC NIST P256 EK Nonce
0x01c0000c	ECC NIST P256 EK Template

EK Certificate NV Index(es) SHOULD be *Populated* by the manufacturer.

The manufacturer SHOULD leave the EK Nonce NV Index *Absent*. See section 2.2.2.2.3.

Note 1: The preferred provisioning uses default EK Templates to conserve TPM NV space.

Note 2: Earlier versions of this specification (V2.1 - V2.3) required that for TPMs designed to meet a specific operating system, the Low Range had to be used for the RSA 2048 EK and the ECC NIST P256 EK. This requirement has been removed from this specification as of version 2.4. The reader needs to consult their respective operating system requirements.

Note 3: The Low Range is for legacy use ONLY; all new EK Certificates and templates defined in this specification are required to use the High Range.

2.2.2.5 High Range

The High Range SHALL use 0x01c00012 and upwards. EK Certificates SHOULD NOT be placed into the range for EK Certificate Chains as defined in Section 2.2.2.5.2.

Any *Populated* even-numbered index MUST contain an EK Certificate. For any EK Certificate, an EK Template MAY be included. If included it MUST be *Populated* in the next (subsequent odd) index.

EK Nonces SHALL NOT be *Populated* in any NV Index in the High Range.

Note 1: There is no need to tightly pack the certificate / template pairs. Any NV Index in the range permitted in the *Registry of Reserved TPM 2.0 Handles and Localities [2]* is acceptable.

If a TCG-defined default template for the EK Certificate exists, the EK Template NV Index SHOULD be *Absent*. If the TCG does not define a default template for the EK Certificate, the EK Template MUST be *Populated*.

Note 2: The preferred provisioning uses default EK Templates to conserve TPM NV space.

2.2.2.5.1 Handle Values for EK Certificates

If EK Certificates are *Populated*, then the following list of NV Index handles SHALL be used to store the EK Certificates corresponding to the EKs created with the default Templates defined in the High Range.

Note 1: Platform profiles define which EK Certificates are required to be *Populated*.

Note 2: The handle values are normative as of version 2.3 of this specification. Defining the NV index handles as normative simplifies EK Certificate validation because it eliminates parsing of certificate content in order to identify the algorithm and key/curve size/parameter set.

Note 3: Handle values in the range 0x01c00020-0x01c0002f are reserved for the Automotive Thin Profile [24].

EK Certificates for Classical Storage EKs:

0x01c00012	RSA 2048 EK Certificate (H-1)
0x01c00014	ECC NIST P256 EK Certificate (H-2)
0x01c00016	ECC NIST P384 EK Certificate (H-3)
0x01c00018	ECC NIST P521 EK Certificate (H-4)
0x01c0001a	ECC SM2_P256 EK Certificate (H-5)
0x01c0001c	RSA 3072 EK Certificate (H-6)
0x01c0001e	RSA 4096 EK Certificate (H-7)
0x01c00040	ECC NIST P256 firmware-limited EK Certificate (H-18)
0x01c00042	ECC NIST P384 firmware-limited EK Certificate (H-19)
0x01c00044	ECC NIST P521 firmware-limited EK Certificate (H-20)
0x01c00046	ECC SM2_P256 firmware-limited EK Certificate (H-21)

EK Certificates for Classical Signing EKs:

0x01c00032	RSA-SSA 2048 EK Certificate (H-8)
0x01c00034	RSA-PSS 2048 EK Certificate (H-9)
0x01c00036	ECC NIST P256 EK Certificate (H-10)
0x01c00038	ECC NIST P384 EK Certificate (H-11)
0x01c0003a	ECC NIST P521 EK Certificate (H-12)
0x01c0003c	ECC SM2_P256 EK Certificate (H-13)
0x01c0003e	RSA-SSA 3072 EK Certificate (H-14)
0x01c00050	RSA-PSS 3072 EK Certificate (H-15)
0x01c00052	RSA-SSA 4096 EK Certificate (H-16)
0x01c00054	RSA-PSS 4096 EK Certificate (H-17)
0x01c00056	ECC NIST P256 firmware-limited EK Certificate (H-22)
0x01c00058	ECC NIST P384 firmware-limited EK Certificate (H-23)
0x01c0005a	ECC NIST P521 firmware-limited EK Certificate (H-24)
0x01c0005c	ECC SM2_P256 firmware-limited EK Certificate (H-25)

EK Certificates for Post-Quantum Storage EKs:

0x01c00060	ML-KEM-512 EK Certificate (H-26)
0x01c00062	ML-KEM-768 EK Certificate (H-27)
0x01c00064	ML-KEM-1024 EK Certificate (H-28)
0x01c00066	ML-KEM-512 firmware-limited EK Certificate (H-29)
0x01c00068	ML-KEM-768 firmware-limited EK Certificate (H-30)
0x01c0006a	ML-KEM-1024 firmware-limited EK Certificate (H-31)

EK Certificates for Post-Quantum Signing EKs:

0x01c00070	ML-DSA-44 EK Certificate (H-32)
0x01c00072	ML-DSA-65 EK Certificate (H-33)
0x01c00074	ML-DSA-87 EK Certificate (H-34)
0x01c00076	ML-DSA-44 firmware-limited EK Certificate (H-35)
0x01c00078	ML-DSA-65 firmware-limited EK Certificate (H-36)
0x01c0007a	ML-DSA-87 firmware-limited EK Certificate (H-37)

2.2.2.5.2 Handle Values for EK Certificate Chains

Storing EK Certificate chains in NVRAM is optional and generally not recommended due to potentially large NV space consumption. Nonetheless, there might be scenarios where provisioning the EK Certificate chain in the TPM is necessary, such as when standard EK Certificate Chain distribution methods are unavailable. If the TPM manufacturer decides to provision the EK Certificate Chain in NV, index handles and formats defined in this section SHOULD be used.

TPM NV MAY contain all certificates of the EK Certificate Chain. The Root CA certificate MUST NOT be contained in NV. The EK Certificate Chain MUST be represented as X.509 DER encoded certificates.

If the TPM manufacturer provisions certificate chains, they MUST be provisioned using the following list of indices:

0x01c00100	EK Certificate Chain Index 1
...	
0x01c001ff	EK Certificate Chain Index 256

The NV indices MUST be *Populated* starting with index 0x01c00100 through index 0x01c001ff. There is no requirement to store the certificates within a chain in any particular order. If a concatenated certificate chain does not fit in a single index, the chain MUST overflow to the next numerically larger index in the list of NV Indices.

If two or more EK Certificate Chains have common certificates, such as when they are anchored to the same intermediate or root CA, the common certificates MUST NOT be stored more than once.

Example 1: a TPM is provided with the following two EK Certificates chains:

1. ECC Root CA -> ECC Intermediate CA 1 -> ECC Issuing CA -> ECC EK Certificate (leaf)
2. RSA Root CA -> RSA Intermediate CA 1 -> RSA Intermediate CA 2 -> RSA Issuing CA -> RSA EK Certificate (leaf)

The EK Certificate Chain NV Indices can be provisioned as follows:

NV Index	Content (... overflow, concatenation)
----------	--

0x01c00100	ECC Issuing CA ECC Intermediate CA 1 ...
0x01c00101	... ECC Intermediate CA 1 RSA Issuing CA RSA Intermediate CA 2 ...
0x01c00102	... RSA Intermediate CA 2 RSA Intermediate CA 1
0x01c00103	(undefined)

In Example 1, the certificate ECC Intermediate CA 1 is too long to store in index 0x01c00100, therefore it overflows into the immediately following index (0x01c00101). The RSA chain is stored immediately following the ECC certificate chain, starting at index 0x01c00101.

Example 2: a TPM is comprised of the following two EK Certificates chains which share the same root and intermediate CA:

1. ECC Root CA -> ECC Intermediate CA -> ECC Issuing CA 1 -> ECC EK Certificate (leaf)
2. ECC Root CA -> ECC Intermediate CA -> ECC Issuing CA 2 -> ECC EK Certificate (leaf)

The EK Certificate Chain NV Indices can be provisioned as follows:

NV Index	Index Content (... overflow, concatenation)
0x01c00100	ECC Issuing CA 1 ECC Intermediate CA ...
0x01c00101	... ECC Intermediate CA ECC Issuing CA 2
0x01c00102	(undefined)

In Example 2, the ECC Intermediate CA certificate is stored only once in NV memory, since the certificate chains share the same intermediate CA.

Example 3: a TPM bootloader includes an embedded CA that issues credentials for both normal and firmware-limited EKs at boot time. The TPM is provided with the following two certificate chains:

1. ECC Root CA -> ECC Intermediate CA -> ECC Issuing CA (embedded within TPM bootloader) -> ECC EK Certificate (leaf)
2. ECC Root CA -> ECC Intermediate CA -> ECC Issuing CA (embedded within TPM bootloader) -> firmware-limited ECC EK Certificate (leaf)

The EK Certificate Chain NV Indices can be provisioned as follows:

NV Index	Index Content (... overflow, concatenation)
0x01c00100	ECC Issuing CA ECC Intermediate CA ...
0x01c00101	... ECC Intermediate CA
0x01c00102	(undefined)

Note 1: It is the responsibility of the TPM vendor to provide mechanisms to perform PKI certificate chain validation. This may involve providing revocation information through a CRL Distribution Point or Online Certificate Status Protocol (OCSP) responder. This specification does not specify mechanisms for CA compromise recovery.

2.2.2.6 TPMT_PUBLIC Calculation

Start of informative comment

This section only applies to Endorsement Keys with certificates in the NV Index Low Range. It does not apply to EKs with certificates in the High Range.

This section is deprecated and may be removed in a future revision.

End of informative comment

The TPMT_PUBLIC structure forms part of the input to the TPM2_CreatePrimary() command.

- If the EK Template is *Absent*, the default template is used as the TPMT_PUBLIC.
- If the EK Template is *Populated* and the EK Nonce is *Absent*, the EK Template is used unmodified as the TPMT_PUBLIC.
- The case of an EK Template *Absent* and an EK Nonce *Populated* is unspecified and MUST NOT be provisioned.
- If the EK Template is *Populated* and the EK Nonce is *Populated*, the TPMT_PUBLIC SHALL be formed as follows:
 1. Begin with the EK Template.
 2. Add the EK Nonce to the default template as follows:
 - For RSA 2048, the EK Nonce is padded to 256 bytes by appending 0x00 bytes. This value is inserted into the default template *unique.rsa.t.buffer*, and *unique.rsa.t.size* is set to 256.
 - For ECC NIST P256, the EK Nonce is padded to 32 bytes by appending 0x00 bytes. This value is inserted into the default template *unique.ecc.x.t.buffer*, and *unique.ecc.x.t.size* is set to 32. The *unique.ecc.y.t.buffer* is set to 32 0x00 bytes and the *unique.ecc.y.size* is set to 32.

2.2.2.7 Locating Keys or specific NV Index Content

Start of informative comment

Persistent Keys and NV Index content typically do not change between platform reset cycles. If locating these entities during a boot cycle is resource sensitive, software should locate Persistent Keys or NV Index content during initial installation or provisioning and store the specific location on the platform.

Discovery of a Persistent Key can be done by TPM2_GetCapability() - TPM_CAP_HANDLES - TPM_HT_PERSISTENT.

Discovery of the NV Index content (for example, a certificate) can be done by TPM2_GetCapability() - TPM_CAP_HANDLES - TPM_HT_NV_INDEX.

End of informative comment

2.2.2.8 Read EK Certificates and create the associated EKs

Start of informative comment

The following describes a high-level procedure how to read the EK Certificates from the TPM and how to create the associated Endorsement Keys and is informative. Differences in the Low and High Range are pointed out.

1. Get a list of all NV indices stored in the handle range reserved for EK Certificates (0x01C00000 – 0x01C07FFF) using TPM2_GetCapability(). The handle range is defined in the Registry of Reserved TPM 2.0 Handles and Localities [2].
2. Identify whether the returned NV index handles lie in the Low Range (0x01C00002 - 0x01C0000C) or in the High Range (0x01C00012 - 0x01C07FFF), excluding the range for the EK Certificate Chain (0x01C00100 – 0x01C001FF).
 - a. In the Low Range, an EK Certificate, an EK Nonce (recommended to be *Absent*), and an EK Template (recommended to be *Absent*) are *Populated* at assigned standard handle values. If present,
 - i. an EK Certificate is at 0x01c00002 (RSA) or 0x01c0000a (ECC)

- ii. an EK Nonce is at 0x01c00003 (RSA) or 0x01c0000b (ECC)
 - iii. an EK Template is at 0x01c00004 (RSA) or 0x01c0000c (ECC)
 - b. In the High Range, an EK Certificate is *Populated* at an even handle value, and (if present) an EK Template is *Populated* at the subsequent odd handle value.
- 3. Read all NV index handles or those of interest returned in the list from step 1 using TPM2_NV_ReadPublic(), and TPM2_NV_Read().
- 4. Identify the type of EK Certificate in order to create the associated EK. See section 2.2.2.5.1 for the mapping of index handle to EK Certificate type.
- 5. Create the associated EK using TPM2_CreatePrimary() (typically with a NULL password)
 - a. Call TPM2_CreatePrimary() with
 - i. *inPublic.publicArea* set to
 - 1. If the EK Template and/or the EK Nonce is *Populated*, the parameter values defined in section 2.2.2.6
 - 2. Otherwise, the parameter values of default EK Templates defined in section 5 of this document
 - ii. *inSensitive.sensitive.data* set to Empty Buffer (see 2.1.1)
 - b. Parse the public key from the returned *outPublic.publicArea.unique* parameter and compare it with the public key stored in the certificate. If they match, the key corresponds to the certificate.

End of informative comment

2.2.3 EK Credential Lifetime

Start of informative comment

An EK Credential contains fields that express the validity period of the credential. The validity period is at the discretion of the manufacturer. The credential is not expected to expire during the normal life expectancy of the platform in which it resides. The lifetime can vary widely between different types of platforms (e.g. while a typical validity period for a PC Client platform is 5-10 years, non-user device TPMs are expected to operate indefinitely into the future in which case the value 99991231235959Z should be used as expiration date). The credential lifetime can also depend on the lifetime of the TPM device and the algorithm type of the Endorsement Key. The time frame during which the security strength of the EK is acceptable needs to be taken into account by the manufacturer when determining the credential lifetime (e.g. see SP800-57 [11]).

However, an EK Credential can become useless before expiration of the validity period if the associated EK is irrevocably erased from the TPM. This is the case if the EPS is replaced (see 2.1.3 EK Lifetime).

TPM 2.0 provides functionality to define a permanent NV Index. In TPM 2.0, the NV Index attributes TPMA_NV_PLATFORMCREATE and TPMA_NV_POLICY_DELETE (see *TPM 2.0 Library Specification, Part 2 [1]*) determine the authorization required to delete an NV Index. If such an NV Index is created such that Platform Authorization is required to write it, the EK Credential can be protected against accidental deletion (e.g. by the Owner).

TPMA_NV_PLATFORMCREATE indicates whether the NV Index was defined by the platform. If SET, the index may only be undefined with Platform Authorization and not with Owner Authorization. TPMA_NV_PLATFORMCREATE SHOULD be SET for an EK Credential to prevent the credential from being deleted if the Owner is cleared. Platform-defined NV indices can also SET TPMA_NV_POLICY_DELETE.

If TPMA_NV_POLICY_DELETE is SET, the Index cannot be deleted unless the *authPolicy* of the NV Index is satisfied using the command TPM2_NV_UndefineSpaceSpecial(). Similarly, it is possible to

create an NV index that cannot be written without a policy authorization. A platform that requires a permanent EK Credential would not create a policy that allows the EK Credential to be removed or overwritten. On the other hand, a platform that wants to clear the EK Credential, e.g. during platform refurbishment, could create a policy that includes among the AND terms the command `TPM2_PolicyCommandCode()` where the command code is set to `TPM_CC_NV_UndefineSpaceSpecial` along with policy commands that control the authorization.

The settings of the NV Index attributes are determined by Platform-specific specifications. Definitions specific to PC Client can be found in *PC Client Specific Platform TPM Profile for TPM 2.0* [6], section Non-volatile Storage.

End of informative comment

2.3 Privacy Protection

In TPM 2.0, privacy-sensitive operations are controlled by the Privacy Administrator. The Privacy Administrator controls the Endorsement Hierarchy and sets the hierarchy authorization and policy (*endorsementAuth* and *endorsementPolicy*). The Privacy Administrator and the Owner are often the same entity.

Because an Endorsement Key is unique to a TPM and usually has a long lifetime, it could be used to identify a user or a platform. Therefore, the EK could be privacy-sensitive. The following applies if protection of privacy is important:

- The usage of the EK is limited by its object attributes, so the EK can only be authorized per its policy (or in some cases by a password authorization, see 5.4). The EK SHOULD be defined with the attributes defined in the appropriate template as documented in section 5 Default EK Templates (algorithm-specific). This prevents the EK from being used for signing operations.

Start of informative comment

The availability of the EK can be controlled with the flag *ehEnable*. The purpose of the flag is to enable and disable the Endorsement Hierarchy. When the Endorsement Hierarchy is disabled (*ehEnable* CLEAR), objects defined under that hierarchy are inaccessible, and *endorsementAuth* and *endorsementPolicy* cannot be used for authorization. It is not possible to use the EK in any command or read the public EK with *TPM2_ReadPublic()*. The *ehEnable* flag may be cleared with the command *TPM2_HierarchyControl()* using Endorsement Authorization or Platform Authorization.

Protection for the EK Credential can be provided by the flag *phEnableNV* if *TPMA_NV_PLATFORMCREATE* is SET in the NV Index attributes of the EK Credential. This attribute indicates whether the index was defined by the platform. When *phEnableNV* is CLEAR, NV space defined by the platform firmware is not accessible, including the EK Credential. This flag can only be cleared by Platform Authorization, which is seldom accessible to the end user.

End of informative comment

3 X.509 ASN.1 Definitions

This section specifies the format for the EK Credential instantiated as an X.509 certificate. All fields are defined in ASN.1 and encoded using DER [19]. The appropriate OIDs are defined in section 4.

Version 3 of the X.509 certificate structure is used for compatibility with existing PKI tools and services. TCG credential profiles do not utilize all aspects of X.509 defined fields and some fields are overloaded with TCG-specific interpretations. The following sections define TCG interpretations for X.509 certificates.

TCG defines a number of new attribute value types to hold TCG-specific values. When present in a public key certificate they are carried in the subject alternative name or subject directory attributes extension.

This specification is a profile of RFC 5280 [12] which is itself a profile of the ISO/IEC/ITU-T X.509 specifications for public key certificates. All syntax and semantics are inherited from those specifications unless explicitly documented otherwise below.

3.1 TCG Attributes

3.1.1 TPM Security Assertions

These attributes describe security-related assertions about the TPM.

Each attribute begins with a version number which identifies the version of the assertion syntax. Future versions of this profile MAY add new assertions by appending new fields at the end of the ASN.1 SEQUENCE and increasing the version number to identify which version of the assertion syntax is encoded.

The **fieldUpgradable** BOOLEAN indicates whether the TPM is capable of having its firmware upgraded after manufacturing.

The **ekGenerationType** indicates how the Endorsement Key in the TPM was created. It MAY be internally generated within the TPM firmware, generated externally and then inserted under a controlled environment during manufacturing, or generated by an entity such as a TPM bootloader that measured TPM firmware and produced a firmware-limited EK bound to those measurements. The revocable variants indicate whether the EK Credential can be revoked or not.

In the **CommonCriteriaMeasures**, the profile and target for the evaluation can be described by either an OID, a URI to a document describing the value, or both. If both are present, they MUST represent consistent values. The URI values are included in an **URIReference** which describes the URI to the document and a cryptographic hash value which identifies a specific version of the document.

URIMAX is a constant used to provide an upper bound on the length of a URI included in the certificate. This upper bound might be helpful to consumers of the extension and also helps limit the overall size of the certificate. In order to provide a reasonable upper bound for ASN.1 parsers, URIMAX SHOULD NOT exceed a value of 1024. This value was selected as it matches the length limit for <A> anchors in HTML as specified by the SGML declaration (LITLEN) for HTML [18].

STRMAX is a constant defining the upper bound on the length of a string type. Like the URIMAX this is to aid ASN.1 parsers and help limit the upper bound on the length of the certificate. Based on the expected sizes of the strings in the ASN.1 in this document an upper bound of 256 was selected. STRMAX SHOULD NOT exceed a value of 256.

```
Version ::= INTEGER { v1(0) }
```

```
tpmSecurityAssertions ATTRIBUTE ::= {
```

```

WITH SYNTAX TPMSecurityAssertions
ID tcg-at-tpmSecurityAssertions }

```

```

TPMSecurityAssertions ::= SEQUENCE {
    version Version DEFAULT v1,
    fieldUpgradable BOOLEAN DEFAULT FALSE,
    ekGenerationType [0] IMPLICIT EKGenerationType OPTIONAL,
    ekGenerationLocation [1] IMPLICIT EKGenerationLocation OPTIONAL,
    ekCertificateGenerationLocation [2] IMPLICIT
        EKCertificateGenerationLocation OPTIONAL,
    ccInfo [3] IMPLICIT CommonCriteriaMeasures OPTIONAL,
    fipsLevel [4] IMPLICIT FIPSLLevel OPTIONAL,
    iso9000Certified [5] IMPLICIT BOOLEAN DEFAULT FALSE,
    iso9000Uri IA5STRING (SIZE (1..URIMAX) OPTIONAL }

```

```

EKGenerationType ::= ENUMERATED {
    internal (0),
    injected (1),
    internalRevocable(2),
    injectedRevocable(3),
    firmwareLimited(4) }

```

```

EKGenerationLocation ::= ENUMERATED {
    tpmManufacturer (0),
    platformManufacturer (1),
    ekCertSigner (2) }

```

```

EKCertificateGenerationLocation ::= ENUMERATED {
    tpmManufacturer (0),
    platformManufacturer (1),
    ekCertSigner (2) }

```

```

-- common criteria evaluation

```

```

CommonCriteriaMeasures ::= SEQUENCE {
    version IA5STRING (SIZE (1..STRMAX)), -- "2.2" or "3.1"; future syntax defined by
    assurancelevel EvaluationAssuranceLevel,
    evaluationStatus EvaluationStatus,
    plus BOOLEAN DEFAULT FALSE,
    strengthOfFunction [0] IMPLICIT StrengthOfFunction OPTIONAL,
    profileOid [1] IMPLICIT OBJECT IDENTIFIER OPTIONAL,
    profileUri [2] IMPLICIT URIReference

```

CC

```

OPTIONAL,
    targetOid [3] IMPLICIT OBJECT IDENTIFIER OPTIONAL,
    targetUri [4] IMPLICIT URIReference OPTIONAL }

```

```

EvaluationAssuranceLevel ::= ENUMERATED {
    level1 (1),
    level2 (2),
    level3 (3),
    level4 (4),
    level5 (5),
    level6 (6),
    level7 (7) }

```

```

StrengthOfFunction ::= ENUMERATED {
    basic (0),
    medium (1),
    high (2) }

```

```

-- Reference to external document containing information relevant to this subject.
-- The hashAlgorithm and hashValue MUST both exist in each reference if either
-- appear at all.

```

```

URIReference ::= SEQUENCE {
    uniformResourceIdentifier IA5String (SIZE (1..URIMAX)),
    hashAlgorithm AlgorithmIdentifier OPTIONAL,
    hashValue BIT STRING OPTIONAL }

```

```

EvaluationStatus ::= ENUMERATED {
    designedToMeet (0),
    evaluationInProgress (1),
    evaluationCompleted (2) }

```

```

-- fips evaluation

```

```

FIPSLLevel ::= SEQUENCE {
    version IA5STRING (SIZE (1..STRMAX)), -- "140-1" or "140-2"
    level SecurityLevel,
    plus BOOLEAN DEFAULT FALSE }

```

```

SecurityLevel ::= ENUMERATED {
    level1 (1),
    level2 (2),
    level3 (3),
    level4 (4) }

```

3.1.2 TPM Device Attributes

The following definitions define the syntax of the relative distinguished names (RDNs) used in the subject alternative name extension to identify the type of the TPM.

The value of the **TPMManufacturer** attribute MUST be the ASCII representation of the hexadecimal value of the 4-byte vendor identifier defined in the TCG *Vendor ID Registry* [3]. Each byte is represented individually as a 2-digit unsigned hexadecimal number using the characters 0-9 and A-F. The result is concatenated together to form an 8-character name which is appended after the lower-case ASCII characters “id:”.

For example, the vendorId 0x12 0x34 0x56 0xEF would be encoded as “id:123456EF”.

Likewise, the value of the **TPMVersion** attribute MUST be the ASCII representation of the hexadecimal value of the 4 bytes derived from the most significant 32 bits of the firmware version of the TPM (TPM_PT_FIRMWARE_VERSION_1). The TPM firmware version is a manufacturer-specific implementation version of the TPM. The version represents the TPM firmware at the time the certificate was created, typically the initial TPM firmware loaded during manufacturing. Each byte is represented individually as a 2-digit unsigned hexadecimal number using the characters 0-9 and A-F. The result is concatenated together to form an 8-character name which is appended after the lower-case ASCII characters “id:”.

For example, a TPM_PT_FIRMWARE_VERSION_1 of 0x00020008 would be encoded as “id:00020008”.

The value of the **TPMModel** attribute is a UTF-8 string that represents the TPM part number. The values are manufacturer-specific.

```
TPMManufacturer ATTRIBUTE ::= {  
    WITH SYNTAX UTF8String (SIZE (1..STRMAX))  
    ID tcg-at-tpmManufacturer }  
  
TPMModel ATTRIBUTE ::= {  
    WITH SYNTAX UTF8String (SIZE (1..STRMAX))  
    ID tcg-at-tpmModel }  
  
TPMVersion ATTRIBUTE ::= {  
    WITH SYNTAX UTF8String (SIZE (1..STRMAX))  
    ID tcg-at-tpmVersion }
```

3.1.3 TPM Specification Attribute

The following definitions define the syntax of the TPM specification attribute.

The **TPMSpecification** attribute identifies the TPM family, level and revision of the TPM specification with which a TPM implementation is compliant. The family value of “2.0” with level 0 and revision 99 identifies a TPM compliant with a public TPM 2.0 specification version 0.99 published by TCG. The family value is encoded in a UTF-8 string but the current defined standard values fall within the ASCII character set.

```
tpmSpecification ATTRIBUTE ::= {  
    WITH SYNTAX TPMSpecification  
    ID tcg-at-tpmSpecification }  
  
TPMSpecification ::= SEQUENCE {
```

```

family UTF8String (SIZE (1..STRMAX)),
level INTEGER,
revision INTEGER }

```

3.1.4 EK Credential Algorithm List Attribute

The following definition defines the syntax of the EK Credential Algorithm List attribute.

The **EKCredentialAlgorithmList** attribute identifies subject public key algorithms to which the Issuer of this credential has issued the EK credential. This can be useful for determining which EK algorithms are available for attestation on a target TPM.

```

TPMEKCredentialAlgorithms ATTRIBUTE ::= {
    WITH SYNTAX EKCredentialAlgorithmList
    ID tcg-at-tpmEKCredentialAlgorithms
}

EKCredentialAlgorithmList ::= SEQUENCE OF OBJECT IDENTIFIER

```

3.1.5 TPM PQC Version Attribute

The following definition defines the syntax of the TPM PQC Version attribute.

The **TPMPQCVersion** attribute identifies the minimum value of the most significant 64 bits of the TPM firmware version (TPM_PT_FIRMWARE_VERSION_1 and TPM_PT_FIRMWARE_VERSION_2) for post-quantum crypto (PQC) support. This can be useful for determining if PQC is expected to be supported on a TPM that shipped with a PQC EK certificate, but required a firmware upgrade in order to use the PQC EK and EK certificate. Note that TPM_PT_FIRMWARE_VERSION_2 can be set to all zeros (see section 3.2.11.3).

```

TPMPQCVersion ATTRIBUTE ::= {
    WITH SYNTAX INTEGER
    ID tcg-at-tpmPQCVersion }

```

3.2 EK Certificate

This section contains the format for a TPM 2.0 EK Credential conforming to this specification. An X.509 EK Certificate is an instantiation of the TPM EK Credential defined in section 2.2.

The “Field Status” column in the table below specifies the presence of the certificate fields. The value “Standard” means the field is an inherent component of the standard certificate syntax and is not optional. The value “MUST”, “SHOULD” or “MAY” is used to indicate the presence of the certificate extensions. The content is described in the “Value” column. Values marked with “(optional)” are added for completeness and are meant to be optional.

Note: This specification does not preclude the use of other certificate extensions. However, any extensions marked as critical will cause interoperability problems when existing clients do not know how to parse the extension and reject it as specified in RFC 5280 [12], section 4.2. (This has historically been a challenge when introducing new critical extensions.)

Field Name	RFC 5280 Type	Value	Field Status
Version	INTEGER	V3 (encoded as value 2)	Standard
Serial Number	INTEGER	Positive integer	Standard
Signature Algorithm	AlgorithmIdentifier	algorithm-specific, see 6.1	Standard
Issuer	Name	Name of issuing CA	Standard

Field Name	RFC 5280 Type	Value	Field Status
Validity	notBefore notAfter	Beginning and end of validity period	Standard
Subject	Name	Unique name assigned by the manufacturer or empty	Standard
Subject Public Key Info	SubjectPublicKeyInfo	algorithm-specific, see 6.2	Standard
Extensions			
Certificate Policies (non-critical)	CertificatePolicies	PolicyIdentifier	MAY
Subject Alternative Name (critical/ non-critical (dep. on subject))	GeneralName directoryName	TPMManufacturer TPMModel TPMVersion	MUST
Basic Constraints (critical)	BasicConstraints	CA=FALSE	MUST
Subject Directory Attributes (non-critical)	SubjectDirectoryAttributes	TPMSpecification Family Level Revision	MAY
		EKCredentialAlgorithmList	MUST if the issuer issued at least one ML-KEM or ML-DSA EK Credential to this TPM; MAY otherwise
		TPMPQCVersion	MUST if PQC support requires a firmware upgrade; MUST NOT otherwise
Authority Key Identifier (non-critical)	AuthorityKeyIdentifier	keyIdentifier authorityCertIssuer (optional) authorityCertSerialNumber (optional)	MUST

Field Name	RFC 5280 Type	Value	Field Status
Authority Info Access (non-critical)	AuthorityInfoAccessSyntax	id-ad-caIssuers URI to issuing CA id-ad-ocsp URI to OCSP responder	MAY
CRL Distribution (non-critical)	CRLDistributionPoints	URI to CRL	MAY
Key Usage (critical)	KeyUsage	keyEncipherment or keyAgreement or digitalSignature	MUST
Extended Key Usage (non-critical)	ExtKeyUsageSyntax	tcg-kp-EKCertificate	MAY
Subject Key Id	SubjectKeyIdentifier	Key identifier	MAY

Table 2: EK Certificate Fields

3.2.1 Version

This field describes the version of the X.509 certificate. Since EK Certificates contain mandatory extensions the version number **MUST** be set to 3 (which is encoded as the value 2 in ASN.1).

3.2.2 Serial Number

The serial number **MUST** be a positive integer which is uniquely assigned to each EK Certificate by the issuer. The combination of an issuer's DN and the serial number **MUST** uniquely describe a single certificate.

3.2.3 Signature Algorithm

This field identifies the algorithm used by the EK Certificate issuer to sign the certificate. This field is algorithm-specific, see section 6.1.

3.2.4 Issuer

This field contains the distinguished name of the certificate issuer which is the entity that vouches that the TPM is genuine and complies with the TPM 2.0 Library Specification [1].

3.2.5 Validity

The period when the certificate is valid is represented by two date values named notBefore and notAfter. Issuers **SHOULD** assign notBefore to the current time when the EK Certificate is issued and notAfter to the last date that the certificate will be considered valid. Both notBefore and notAfter **MUST** use the appropriate time format as indicated by RFC 5280 [12]. (See also section 2.2.3 EK Credential Lifetime)

3.2.6 Subject

The subject field **MUST** contain an X.500 distinguished name (DN) that uniquely identifies the TPM or, if unique identification through the subject field is not required, **MUST** be empty.

If the subject name field is empty, the subject alternative name extension **MUST** be critical in accordance with RFC 5280 [12], otherwise it **SHOULD** be non-critical.

The subject field **MAY** contain a device (TPM) serial number in the attribute id-at-serialNumber.

Note: The TPM serial number usually includes detailed production parameters. Since that might reveal information that the manufacturer does not want to disclose, the hash of the TPM serial number could be used instead in the EK Credential.

3.2.7 Subject Public Key Info

This describes the public Endorsement Key algorithm and key value. This field is algorithm-specific, see section 6.2.

3.2.8 Certificate Policies

This extension indicates the policy terms under which the certificate was issued. This extension is optional. If included, it SHOULD be non-critical and PolicyIdentifier MUST have at least one object identifier. Policy qualifiers, such as the cPSuri policy qualifier and the userNotice policy qualifier SHOULD NOT be included.

Note: A pointer to a Certification Practice Statement (CPS) can be provided in the manufacturer's datasheet instead.

3.2.9 Subject Alternative Name

This contains the alternative name of the entity associated with this certificate. The issuer MUST include TPM manufacturer, TPM part number and TPM firmware version, using the directoryName-form within the GeneralName structure. The ASN.1 encoding is specified in section 3.1.2 TPM Device Attributes. In accordance with RFC 5280 [12], this extension MUST be critical if the subject is empty and SHOULD be non-critical if subject is non-empty.

- The TPM manufacturer identifies the manufacturer of the TPM. This value MUST be the vendor ID defined in the TCG *Vendor ID Registry* [3]. It MUST match the value reported by the command TPM2_GetCapability(property = TPM_PT_MANUFACTURER).
- The TPM part number is encoded as a string and is manufacturer-specific. A manufacturer MUST provide a way for the user to retrieve the part number physically or logically. This information could be e.g. provided as part of the vendor string in the command TPM2_GetCapability(property = TPM_PT_VENDOR_STRING_x; x=1...4).
- The TPM firmware version is a manufacturer-specific implementation version of the TPM. The version represents the TPM firmware at the time the certificate was created, typically the initial TPM firmware loaded during manufacturing. In the case of a Field Upgrade, the firmware version will change and this value will no longer match the version reported by the command TPM2_GetCapability (property = TPM_PT_FIRMWARE_VERSION_1).

Version 2.0 of this specification [5] allowed the inclusion of an optional attribute, called HardwareModuleName that contains the TPM serial number. HardwareModuleName SHOULD NOT be used in EK Certificates conforming to this version of this specification. Instead, if a serial number is present, it SHOULD be stored in the subject field (see 3.2.6).

3.2.10 Basic Constraints

This extension indicates whether the subject is a CA. "CA" MUST be set to FALSE. This extension MUST be critical.

3.2.11 Subject Directory Attributes

The extension includes miscellaneous properties and security assertions about the entity. This extension MUST be non-critical.

3.2.11.1 TPM Specification Attribute

The following attribute MAY be included in a subject directory attributes extension in the EK Certificate:

- The “TPM Specification” attribute that identifies the family, level and revision of the TCG TPM specification to which the TPM was designed. The ASN.1 encoding is specified in section 3.1.3 TPM Specification Attribute.

Version 2.0 of this specification [5] allowed the inclusion of an optional attribute, called “TPM Security Assertions”, which described various assertions about the security properties of the TPM and the conditions under which the Endorsement Key was generated. The Security Assertions attribute SHOULD NOT be included in EK Certificates conforming to this version of this specification.

3.2.11.2 EK Credential Algorithm List Attribute

Start of informative comment

To help verifiers and other relying parties to detect downgrade attacks (e.g., some interposer drops the “best” EK certificate from a request, forcing the verifier to use a “worse” one), some EK Credentials are required to contain an EKCredentialAlgorithmList attribute containing a list of EK algorithm OIDs (indicating which EK Credentials have been issued by the same Issuer). Each OID identifies the EK algorithm or parameter set associated with an EK according to Table 3.

End of informative comment

The EKCredentialAlgorithmList attribute is REQUIRED when the EK credential Issuer issued at least one ML-KEM or ML-DSA EK Credential. It is OPTIONAL in other cases.

The EKCredentialAlgorithmList attribute (when present) MUST contain an OID for the largest (by target security strength) issued EK Credential algorithm/parameter set in each algorithm family from Table 3.

If there are multiple EK credentials that share the same algorithm/parameter OID (e.g., a signing EK and a storage EK that use the same ECC curve), the OID SHALL be listed only once in the EKCredentialAlgorithmList.

Start of informative comment

It is possible for a given EK Credential for some algorithm to have been issued, but not be currently available (e.g., because the TPM requires a firmware upgrade in order to use a PQC EK credential). The EKCredentialAlgorithmList is still required in this case.

The TPMPQCVersion attribute (see 3.2.11.3) informs the Verifier that this might have been the case, allowing the Verifier to check the TPM’s current firmware version and decide whether to e.g., allow an exception or request that the TPM firmware be upgraded.

End of informative comment

Example 1: A TPM with EK Credentials for RSA-2048, ECC NIST-P256 and P384, and ML-KEM-768 and -1024 will have an EKCredentialAlgorithmList attribute with OIDs for rsaEncryption, secp384r1 and ml-kem-1024 in each of its EK Credentials.

Example 2: A TPM with EK Credentials for NIST-P384 (storage as well as signing), and ML-KEM1024 and ML-DSA87 will have an EKCredentialAlgorithmList attribute with OIDs for ml-kem-1024, ml-dsa-87, and secp384r1 in each of its EK Credentials.

Example 3: A TPM with EK Credentials for RSA-3072 and ECC NIST P-384 might have an EKCredentialAlgorithmList attribute with OIDs for secp384r1 and rsaEncryption in each of its EK Credentials.

Table 3 Certificate Policy EK Algorithm Identifiers

Family	EK Algorithm/Parameter Set	OID
RSA	RSA (any size)	1.2.840.113549.1.1.1 (rsaEncryption), specified in RFC 8017

Family	EK Algorithm/Parameter Set	OID
ECC (NIST Curves)	ECC NIST P-256	1.2.840.10045.3.1.7 (secp256r1), specified in RFC 5480
	ECC NIST P-384	1.3.132.0.34 (secp384r1), specified in RFC 5480
	ECC NIST P-521	1.3.132.0.35 (secp521r1), specified in RFC 5480
ECC (SM2)	ECC SM2 P256	1.2.156.10197.1.301 (SM2EllipticCurveCryptography), specified in GM/T 0006-2012
ML-KEM	ML-KEM-512	2.16.840.1.101.3.4.4.1 (ml-kem-512), specified in draft-ietf-lamps-kyber-certificates
	ML-KEM-768	2.16.840.1.101.3.4.4.2 (ml-kem-768), specified in draft-ietf-lamps-kyber-certificates
	ML-KEM-1024	2.16.840.1.101.3.4.4.3 (ml-kem-1024), specified in draft-ietf-lamps-kyber-certificates
ML-DSA	ML-DSA-44	2.16.840.1.101.3.4.3.17 (ml-dsa-44), specified in draft-ietf-lamps-dilithium-certificates
	ML-DSA-65	2.16.840.1.101.3.4.3.18 (ml-dsa-65), specified in draft-ietf-lamps-dilithium-certificates
	ML-DSA-87	2.16.840.1.101.3.4.3.19 (ml-dsa-87), specified in draft-ietf-lamps-dilithium-certificates

3.2.11.3 TPM PQC Version Attribute

If the TPM requires a post-manufacturing firmware upgrade in order to use its ML-KEM or ML-DSA EK Credential, then the TPMPQCVersion attribute MUST be present. Otherwise, it MUST NOT be present. If present, the TPMPQCVersion MUST contain a value that, when interpreted as a 64-bit big-endian integer, is

- greater than the value of TPM_PT_FIRMWARE_VERSION_1 and TPM_PT_FIRMWARE_VERSION_2 for the TPM firmware version at the time of manufacturing and all future TPM firmware versions that do not support the use of its ML-KEM or ML-DSA EK Credentials
- less than or equal to the value of TPM_PT_FIRMWARE_VERSION_1 and TPM_PT_FIRMWARE_VERSION_2 for all future TPM firmware versions that enabled the use of its ML-KEM or ML-DSA EK Credentials

TPM_PT_FIRMWARE_VERSION_2 in the TPMPQCVersion attribute MAY be set to all zeros if TPM_PT_FIRMWARE_VERSION_1 sufficiently identifies the TPM firmware version.

Start of informative comment

This allows a verifier to numerically compare the TPM's current firmware version and the version encoded in an EK certificate's TPMPQCVersion attribute to conclude whether PQC is currently supported on a TPM.

End of informative comment

3.2.12 Authority Key Identifier

This identifies the subject public key of the certificate issuer and hence facilitates the validation of the certificate path. The certificate MUST contain an AuthorityKeyIdentifier that matches the subject key identifier of the CA certificate. The issuer name and the serial number are optional. This extension MUST be non-critical.

3.2.13 Authority Information Access

This extension provides additional information about the issuer. Authority Information Access MAY contain the accessMethod OID id-ad-calssuers (see note 1 below) and/ or the OID id-ad-ocsp. This extension MUST be non-critical.

If id-ad-calssuers appears as accessMethod, then the accessLocation value SHOULD point to the URL where the certificate of the issuing CA can be retrieved.

Note: The root certificate should not be retrieved from a URL in the AIA extension as this is a possible attack vector. Instead, the root certificate should be provisioned in a trusted root store out of band or from a trusted source. Alternatively, a hash of the root public key or certificate could be stored in a device root of trust.

If id-ac-ocsp appears as accessMethod, then the accessLocation value SHOULD point to the access value of the OCSP responder (HTTP URI). The relying party can access the certificate status for this certificate by sending a properly formatted OCSPRequest to the URI. If both a CDP and OCSP AIA extension are present in the certificate, then the relying parties SHOULD use OCSP as the primary validation mechanism.

3.2.14 CRL Distribution

This extension is optional and provides the location of the subject's revocation information. The relying party can access the CRL for this certificate from this URI. If both a CDP and OCSP AIA extension are present in the certificate, then relying parties SHOULD use OCSP as the primary validation mechanism. This extension MUST be non-critical.

3.2.15 Key Usage

This extension indicates the intended purpose of the subject public key. This extension MUST be critical. The usage of the Endorsement Key is defined by the object attributes and algorithm in its public area template.

If the EK has the object attributes

- *restricted, decrypt* SET (i.e. the EK is a Storage key),
 - the keyEncipherment bit MUST be set for an RSA EK Certificate
 - the keyAgreement bit MUST be set for an ECC EK Certificate
 - the keyEncipherment bit MUST be set for an ML-KEM EK Certificate
- *restricted, sign* SET (i.e. the EK is a Signing key), the digitalSignature bit MUST be set

3.2.16 Extended Key Usage

The extended key usage (EKU) extension indicates the intended purpose of the subject public key and MAY be included in the EK Certificate (see note 1 below). If present, extended key usage SHOULD contain the OID tcg-kp-EKCertificate defined in section 5 of this document as shown below. The OID is used to unambiguously identify the certificate as an EK Certificate. This extension MUST be non-critical.

```
tcg-kp-EKCertificate OBJECT IDENTIFIER ::= {  
    joint-iso-itu-t(2) international-organizations(23) tcg(133) kp(8) 1}
```

Note: If the issuing CA is used exclusively to issue EK certificates, the OID `tcg-kp-EKCertificate` MAY also be included in the issuing CA certificate. This ensures that the use of the CA is limited to that particular purpose. If the issuing CA issues certificates for multiple known purposes, then the set of relevant EKU OIDs could be included in the issuing CA certificate.

3.2.17 Subject Key Identifier

This identifies the public key of the certificate. This extension MAY be included. If included, it MUST be non-critical.

4 X.509 ASN.1 Structures and OIDs

TCG has registered an object identifier (OID) namespace as an “international body” in the ISO registration hierarchy. This leads to shorter OIDs and gives TCG the ability to manage its own namespace. The OID namespace is inherited from the Trusted Computing Platform Alliance. These definitions are intended to be used within the context of an X.509 v3 certificate specifically leveraging the profile described in RFC 5280 [12].

```
-- TCG specific OIDs
tcg OBJECT IDENTIFIER ::= {
    joint-iso-itu-t(2) international-organizations(23) tcg(133) }

tcg-attribute OBJECT IDENTIFIER ::= {tcg 2}
tcg-kp OBJECT IDENTIFIER ::= {tcg 8}

-- TCG Attribute OIDs
tcg-at-tpmManufacturer OBJECT IDENTIFIER ::= {tcg-attribute 1}
tcg-at-tpmModel OBJECT IDENTIFIER ::= {tcg-attribute 2}
tcg-at-tpmVersion OBJECT IDENTIFIER ::= {tcg-attribute 3}
tcg-at-tpmSpecification OBJECT IDENTIFIER ::= {tcg-attribute 16}
tcg-at-tpmSecurityAssertions OBJECT IDENTIFIER ::= {tcg-attribute 18}
tcg-at-tpmEKCredentialAlgorithms OBJECT IDENTIFIER ::= {tcg-attribute 26}
tcg-at-tpmPQCVersion OBJECT IDENTIFIER ::= {tcg-attribute 27}

-- TCG Key Purposes OIDs
tcg-kp-EKCertificate OBJECT IDENTIFIER ::= {tcg-kp 1}

-- tcg specification attributes for tpm
tpmSpecification ATTRIBUTE ::= {
    WITH SYNTAX TPMSpecification
    ID tcg-at-tpmSpecification }

TPMSpecification ::= SEQUENCE {
    family UTF8String (SIZE (1..STRMAX)),
    level INTEGER,
    revision INTEGER }

-- manufacturer implementation model and version attributes
TPMManufacturer ATTRIBUTE ::= {
    WITH SYNTAX UTF8String (SIZE (1..STRMAX))
    ID tcg-at-tpmManufacturer }

TPMModel ATTRIBUTE ::= {
    WITH SYNTAX UTF8String (SIZE (1..STRMAX))
    ID tcg-at-tpmModel }

TPMVersion ATTRIBUTE ::= {
    WITH SYNTAX UTF8String (SIZE (1..STRMAX))
    ID tcg-at-tpmVersion }

-- tpm security assertions
Version ::= INTEGER { v1(0) }

tpmSecurityAssertions ATTRIBUTE ::= {
    WITH SYNTAX TPMSecurityAssertions
    ID tcg-at-tpmSecurityAssertions
}

TPMSecurityAssertions ::= SEQUENCE {
    version Version DEFAULT v1,
```

```

        fieldUpgradable BOOLEAN DEFAULT FALSE,
        ekGenerationType [0] IMPLICIT EKGenerationType OPTIONAL,
        ekGenerationLocation [1] IMPLICIT EKGenerationLocation OPTIONAL,
        ekCertificateGenerationLocation [2] IMPLICIT
            EKCertificateGenerationLocation OPTIONAL,
        ccInfo [3] IMPLICIT CommonCriteriaMeasures OPTIONAL,
        fipsLevel [4] IMPLICIT FIPSLevel OPTIONAL,
        iso9000Certified [5] IMPLICIT BOOLEAN DEFAULT FALSE,
        iso9000Uri IA5STRING (SIZE (1..URIMAX)) OPTIONAL }

EKGenerationType ::= ENUMERATED {
    internal (0),
    injected (1),
    internalRevocable(2),
    injectedRevocable(3),
    firmwareLimited(4) }

EKGenerationLocation ::= ENUMERATED {
    tpmManufacturer (0),
    platformManufacturer (1),
    ekCertSigner (2) }

EKCertificateGenerationLocation ::= ENUMERATED {
    tpmManufacturer (0),
    platformManufacturer (1),
    ekCertSigner (2) }

-- common criteria evaluation
CommonCriteriaMeasures ::= SEQUENCE {
    version IA5STRING (SIZE (1..STRMAX)), -- "2.2" or "3.1"; future syntax defined by
    assurancelevel EvaluationAssuranceLevel,
    evaluationStatus EvaluationStatus,
    plus BOOLEAN DEFAULT FALSE,
    strengthOfFunction [0] IMPLICIT StrengthOfFunction OPTIONAL,
    profileOid [1] IMPLICIT OBJECT IDENTIFIER OPTIONAL,
    profileUri [2] IMPLICIT URIReference OPTIONAL,
    targetOid [3] IMPLICIT OBJECT IDENTIFIER OPTIONAL,
    targetUri [4] IMPLICIT URIReference OPTIONAL }

EvaluationAssuranceLevel ::= ENUMERATED {
    level1 (1),
    level2 (2),
    level3 (3),
    level4 (4),
    level5 (5),
    level6 (6),
    level7 (7) }

StrengthOfFunction ::= ENUMERATED {
    basic (0),
    medium (1),
    high (2) }

URIReference ::= SEQUENCE {
    uniformResourceIdentifier IA5String (SIZE (1..URIMAX)),
    hashAlgorithm AlgorithmIdentifier OPTIONAL,
    hashValue BIT STRING OPTIONAL }

EvaluationStatus ::= ENUMERATED {
    designedToMeet (0),
    evaluationInProgress (1),
    evaluationCompleted (2) }

-- fips evaluation

```



```
FIPSLLevel ::= SEQUENCE {  
    version IA5STRING (SIZE (1..STRMAX)), -- "140-1" or "140-2"  
    level SecurityLevel,  
    plus BOOLEAN DEFAULT FALSE }  
  
SecurityLevel ::= ENUMERATED {  
    level1 (1),  
    level2 (2),  
    level3 (3),  
    level4 (4) }
```

5 Default EK Templates (algorithm-specific)

5.1 Introduction

This section defines multiple default EK Templates for algorithm-specific Endorsement Keys. The Templates are assigned to either the Low or High Range. These default values are used to generate the Endorsement Key corresponding to the EK Certificates installed by the TPM or Platform manufacturer. The hash calculated over the public area template is one of the command parameters that is used to create the EK.

The TPM or Platform manufacturer MAY create a proprietary EK public area template that is different from the defaults in this section (this procedure is further described in section 2.2.2). Platform-specific working groups MAY define a proprietary EK public area template if required. The EK provided by the manufacturer MUST be defined as a non-duplicable key. This is ensured by setting the attributes *fixedTPM* and *fixedParent* both to SET (1). The *noDA* attribute is set to CLEAR (0) (meaning the object is subject to dictionary attack protections) in the default EK Templates because the EK is considered privacy-sensitive. If not following the templates defined in this section and privacy is a concern, caution is needed when selecting the attributes and the authorization for the key (see section 2.3 privacy protection).

5.2 Backwards Compatibility

Version 2.0 of the EK Credential Profile [5] defined one EK Template for RSA 2048 and one for ECC NIST P256. These Templates are retained unmodified in this version of the EK Credential Profile and are named L-1 (see 5.3.3) and L-2 (see 5.3.4). In addition, a second EK Template for RSA 2048 and ECC NIST P256 are defined, which utilize a different authorization.

If a platform-specific TPM profile requires an EK Certificate from the manufacturer for RSA 2048 and/or ECC NIST P256, the platform specification SHOULD specify which EK Template is to be used to create the associated EK. For backwards compatibility, it is recommended that TPMs use the Templates L-1 and L-2 in the Low Range instead of H-1 and H-2 in the High Range.

5.3 EK Templates in the Low Range

5.3.1 Introduction

The EK Certificates associated with the default EK Templates defined in this section are stored in the Low Range (see section 2.2.2.4). The default EK Templates L-1 and L-2 (see sections 5.3.3 and 5.3.4) specify an RSA 2048 bit and ECC NIST P256 bit restricted decryption (Storage) key whose authorization is only allowed with *authPolicy*. The policies in Template L-1 and L-2 require *endorsementAuth* to authorize use of the EK. This policy is created with `TPM2_PolicySecret()` where the *authHandle* parameter of the command, indicating the entity providing the authorization value, references the Endorsement Hierarchy. In the following, this policy is referenced as *PolicyA*. A symmetric key is used to protect the child keys of the EK and is defined as an AES 128-bit key using CFB mode. SHA256 is used to calculate the Name of the EK. The buffer reserved for the public key of the EK is set to all zeros. For the setting of the sensitive data, see section 2.1.1.

Using *PolicyA* rather than a policy using the EK authorization (*userWithAuth* SET) enables the following use case: in the case an entity pre-provisions an EK in persistent storage; since the password of a persistent object cannot be changed, the entity needs to:

- communicate the password to the final user, and
- securely delete the copy of the password held by the entity

This policy (*PolicyA*) used by the EK Templates in the Low Range permits the EK to be pre-provisioned while letting the end user set its password (through endorsement authorization).

The disadvantage of *PolicyA* is that it results in reduced flexibility for the Privacy Administrator when delegating control of the EK. If an administrator wishes to give an end user the ability to use the EK,

the administrator has to give the end user the Endorsement Hierarchy password. Among other things, knowledge of the EH password allows the end user to change the EH password and policy, potentially locking out the administrator from using the EK.

5.3.2 Satisfying *PolicyA*

When using the EK created with Template L-1 or L-2, the user has to satisfy *PolicyA*. This is done by executing the TPM2_PolicySecret() command, passing in the handle to the Endorsement Hierarchy, and then proving knowledge of the Endorsement Hierarchy password. The caller proves knowledge of the Endorsement Hierarchy password using an authorization session. A password session, an HMAC session, or a policy session containing TPM2_PolicyAuthValue() or TPM2_PolicyPassword() will satisfy this requirement.

5.3.3 Template L-1: RSA 2048 (Storage)

Parameter	Type	Content
type	TPMI_ALG_PUBLIC	TPM_ALG_RSA
nameAlg	TPMI_ALG_HASH	TPM_ALG_SHA256
objectAttributes	TPMA_OBJECT	fixedTPM = 1 stClear = 0 fixedParent = 1 sensitiveDataOrigin = 1 userWithAuth = 0 adminWithPolicy = 1 firmwareLimited = 0 svnLimited = 0 noDA = 0 encryptedDuplication = 0 restricted = 1 decrypt = 1 sign = 0
authPolicy	TPM2B_DIGEST	
size	UINT16	32
buffer	BYTE	0x83, 0x71, 0x97, 0x67, 0x44, 0x84, 0xB3, 0xF8, 0x1A, 0x90, 0xCC, 0x8D, 0x46, 0xA5, 0xD7, 0x24, 0xFD, 0x52, 0xD7, 0x6E, 0x06, 0x52, 0x0B, 0x64, 0xF2, 0xA1, 0xDA, 0x1B, 0x33, 0x14, 0x69, 0xAA TPM2_PolicySecret(TPM_RH_ENDO RSEMENT)
parameters	TPMS_RSA_PARMS	
symmetric->algorithm	TPMI_ALG_SYM_OBJECT	TPM_ALG_AES
symmetric->keyBits	TPMI_AES_KEY_BITS	128
symmetric->mode	TPMI_ALG_SYM_MODE	TPM_ALG_CFB
scheme->scheme	TPMI_ALG_ASYM_SCHEME	TPM_ALG_NULL
keyBits	TPMI_RSA_KEY_BITS	2048
exponent	UINT32	0
unique	TPM2B_PUBLIC_KEY_RSA	
size	UINT16	256
buffer	BYTE	All 0

Table 4: Default EK Template (TPMT_PUBLIC) L-1: RSA 2048 (Storage)

5.3.4 Template L-2: ECC NIST P256 (Storage)

Parameter	Type	Content
type	TPMI_ALG_PUBLIC	TPM_ALG_ECC
nameAlg	TPMI_ALG_HASH	TPM_ALG_SHA256
objectAttributes	TPMA_OBJECT	fixedTPM = 1 stClear = 0 fixedParent = 1 sensitiveDataOrigin = 1 userWithAuth = 0 adminWithPolicy = 1 firmwareLimited = 0 svnLimited = 0 noDA = 0 encryptedDuplication = 0 restricted = 1 decrypt = 1 sign = 0
authPolicy	TPM2B_DIGEST	
size	UINT16	32
buffer	BYTE	0x83, 0x71, 0x97, 0x67, 0x44, 0x84, 0xB3, 0xF8, 0x1A, 0x90, 0xCC, 0x8D, 0x46, 0xA5, 0xD7, 0x24, 0xFD, 0x52, 0xD7, 0x6E, 0x06, 0x52, 0x0B, 0x64, 0xF2, 0xA1, 0xDA, 0x1B, 0x33, 0x14, 0x69, 0xAA TPM2_PolicySecret(TPM_RH_ENDO RSEMENT)
parameters	TPMS_ECC_PARMS	
symmetric->algorithm	TPMI_ALG_SYM_OBJECT	TPM_ALG_AES
symmetric->keyBits	TPMI_AES_KEY_BITS	128
symmetric->mode	TPMI_SYM_MODE	TPM_ALG_CFB
scheme->scheme	TPMI_ALG_ECC_SCHEME	TPM_ALG_NULL
curveID	TPMI_ECC_CURVE	TPM_ECC_NIST_P256
kdf->scheme	TPMI_ALG_KDF	TPM_ALG_NULL
unique	TPMS_ECC_POINT	
x->size	UINT16	32
x->buffer	BYTE	All 0
y->size	UINT16	32
y->buffer	BYTE	All 0

Table 5: Default EK Template (TPMT_PUBLIC) L-2: ECC NIST P256 (Storage)

5.4 EK Templates in the High Range

5.4.1 Introduction

The EK Certificates associated with the default EK Templates defined in this section are stored in the High Range (see section 2.2.2.5). The default EK Templates H-x (see 5.4.6) specify either a restricted decryption (Storage) key or a restricted signing key whose authorization is allowed with either the object's *authValue* (as *userWithAuth* is SET) or the object's *authPolicy*. The *authPolicy* in the high-range EK Templates is set to *PolicyB*.

PolicyB is a policy OR term of

- *PolicyA*: a policy created with `TPM2_PolicySecret()` where the entity providing the authorization value is the Endorsement Hierarchy, and
- *PolicyC*: a policy created with `TPM2_PolicyAuthorizeNV()` where the authorization value is determined by the data (payload) area of a reserved NV Index.

A symmetric key is used to protect the child keys of the EK and is defined as an AES 128-bit, AES 256-bit, or SM4 128-bit key using CFB mode. The hash algorithm used to calculate the Name of the EK is defined as SHA256, SHA384, SHA512, or SM3_256 and has a security strength equivalent to the Endorsement Key. The *unique* field (reserved for the public key of the EK) is set to Empty Buffer for the RSA, ML-KEM and ML-DSA default EK Templates, and set to Empty Point for the ECC default EK Templates. For the setting of the sensitive data, see section 2.1.1.

5.4.2 Authorization Options

The authorization of the default EK Templates defined in the High Range (H-x) is improved in two ways (compared to the Low Range):

- 1) The policy of the EK is changed to an OR policy that allows authorization of the EK with either:
 - a. The EH password, or
 - b. A policy (only writeable with knowledge of the EH password) at a particular NV index (see A.1).
- 2) *userWithAuth* is SET (1) so that the EK can also be used by someone with knowledge of the EK's *authValue*.

The policy in the default High Range EK Templates allows entity that knows the EH *authValue* to assign (or change) a policy to use the EK. In this case, the entity needs to create an NV index (at a reserved index) and write the policy to the NV Index. Otherwise, no additional NV indexes are used.

The problem of an entity creating an EK with a known *authValue* and storing it persistently in the TPM before the EH password is assigned using `TPM2_HierarchyChangeAuth()` can be obviated in a few ways, by either:

- 1) Clearing the TPM using `TPM2_Clear()` and then changing the EH password when it is recreated, or
- 2) Changing the EH password and then checking with the TPM to see if any keys are persistently stored and evicting them.

After either of these operations, only someone with EH authority is able to create a new EK, assigning it a new password.

5.4.3 Satisfying *PolicyB*

When using the EK created with one of the Templates in the High Range (H-x), the user has to either prove knowledge of the object's *authValue* or satisfy *PolicyB*. *PolicyB* can be satisfied by either satisfying *PolicyA* (see 5.3.2) OR by satisfying *PolicyC*. *PolicyC* is stored in a reserved NV index (defined in A.1). The NV index is established with a policy that only allows it to be written by the

Endorsement Hierarchy owner. The Name of this NV index (after it is written by the Endorsement Hierarchy owner), is used in creation of *PolicyC*.

When the TPM is shipped, the NV Index will not be *Populated* as described in A.1.1. It is entirely up to the owner of the TPM to populate it. If it is not *Populated*, the EK is controlled with the Endorsement Hierarchy authorization. If the owner decides to populate the NV Index with an owner-chosen policy, the EK can be controlled either with that policy OR with the Endorsement Hierarchy authorization. In this case, the owner can now delegate control of the EK via use of *PolicyB*. *PolicyB* can be satisfied either by first executing *PolicyA* and then executing TPM2_PolicyOR(). It can also be satisfied by first satisfying whatever policy was placed in the reserved NV Index, then executing TPM2_PolicyAuthorizeNV(), and then executing TPM2_PolicyOR().

5.4.4 High Range EK Template Overview

Table 6 provides an overview of the EK Templates in the High Range specified in this section.

Table 6 List of EK Templates and Handles

EK Templates	H-x Reference used in section 2.2.2.5.1
RSA Storage (see 5.4.6.1)	
RSA 2048 Storage EK Certificate	H-1
RSA 3072 Storage EK Certificate	H-6
RSA 4096 Storage EK Certificate	H-7
RSA Signing (see 5.4.6.2)	
RSA-SSA 2048 Signing EK Certificate	H-8
RSA-PSS 2048 Signing EK Certificate	H-9
RSA-SSA 3072 Signing EK Certificate	H-14
RSA-PSS 3072 Signing EK Certificate	H-15
RSA-SSA 4096 Signing EK Certificate	H-16
RSA-PSS 4096 Signing EK Certificate	H-17
ECC Storage (see 5.4.6.3)	
ECC NIST P256 Storage EK Certificate	H-2
ECC NIST P384 Storage EK Certificate	H-3
ECC NIST P521 Storage EK Certificate	H-4
ECC SM2_P256 Storage EK Certificate	H-5
ECC NIST P256 firmware-limited Storage EK Certificate	H-18
ECC NIST P384 firmware-limited Storage EK Certificate	H-19
ECC NIST P521 firmware-limited Storage EK Certificate	H-20
ECC SM2_P256 firmware-limited Storage EK Certificate	H-21
ECC Signing (see 5.4.6.4)	
ECC NIST P256 Signing EK Certificate	H-10
ECC NIST P384 Signing EK Certificate	H-11
ECC NIST P521 Signing EK Certificate	H-12
ECC SM2_P256 Signing EK Certificate	H-13
ECC NIST P256 firmware-limited Signing EK Certificate	H-22
ECC NIST P384 firmware-limited Signing EK Certificate	H-23
ECC NIST P521 firmware-limited Signing EK Certificate	H-24
ECC SM2_P256 firmware-limited Signing EK Certificate	H-25
ML-KEM Storage (see 5.4.6.5)	
ML-KEM-512 Storage EK Certificate	H-26
ML-KEM-768 Storage EK Certificate	H-27
ML-KEM-1024 Storage EK Certificate	H-28
ML-KEM-512 firmware-limited Storage EK Certificate	H-29
ML-KEM-768 firmware-limited Storage EK Certificate	H-30
ML-KEM-1024 firmware-limited Storage EK Certificate	H-31
ML-DSA Signing (see 5.4.6.6)	
ML-DSA-44 Signing EK Certificate	H-32
ML-DSA-65 Signing EK Certificate	H-33
ML-DSA-87 Signing EK Certificate	H-34
ML-DSA-44 firmware-limited Signing EK Certificate	H-35
ML-DSA-65 firmware-limited Signing EK Certificate	H-36
ML-DSA-87 firmware-limited Signing EK Certificate	H-37

5.4.5 Common Definitions

5.4.5.1 Object Attributes

Table 7 specifies the object attributes for storage and signing EKs (with and without *firmwareLimited* SET) that are used by the EK Templates in the High Range.

Table 7 Object Attributes for EK Templates

Bit	Object attribute	Storage (attributes- storage)	Signing (attributes- signing)	Firmware- limited Storage (attributes-fw- storage)	Firmware- limited Signing (attributes-fw- signing)
1	fixedTPM	1	1	1	1
2	stClear	0	0	0	0
4	fixedParent	1	1	1	1
5	sensitiveDataOrigin	1	1	1	1
6	userWithAuth	1	1	1	1
7	adminWithPolicy	1	1	1	1
8	firmwareLimited	0	0	1	1
9	svnLimited	0	0	0	0
10	noDA	0	0	0	0
11	encryptedDuplication	0	0	0	0
16	restricted	1	1	1	1
17	decrypt	1	0	1	0
18	sign	0	1	0	1
19	x509sign	0	0	0	0

5.4.5.2 PolicyB Values

Table 8 specifies the *PolicyB* values computed with different hash algorithms that are used by the EK Templates in the High Range as *authPolicy*.

Table 8 PolicyB Values for EK Templates

	Values
PolicyB_{SHA256}	0xCA, 0x3D, 0x0A, 0x99, 0xA2, 0xB9, 0x39, 0x06, 0xF7, 0xA3, 0x34, 0x24, 0x14, 0xEF, 0xCF, 0xB3, 0xA3, 0x85, 0xD4, 0x4C, 0xD1, 0xFD, 0x45, 0x90, 0x89, 0xD1, 0x9B, 0x50, 0x71, 0xC0, 0xB7, 0xA0
PolicyB_{SHA384}	0xB2, 0x6E, 0x7D, 0x28, 0xD1, 0x1A, 0x50, 0xBC, 0x53, 0xD8, 0x82, 0xBC, 0xF5, 0xFD, 0x3A, 0x1A, 0x07, 0x41, 0x48, 0xBB, 0x35, 0xD3, 0xB4, 0xE4, 0xCB, 0x1C, 0x0A, 0xD9, 0xBD, 0xE4, 0x19, 0xCA, 0xCB, 0x47, 0xBA, 0x09, 0x69, 0x96, 0x46, 0x15, 0x0F, 0x9F, 0xC0, 0x00, 0xF3, 0xF8, 0x0E, 0x12
PolicyB_{SHA512}	0xB8, 0x22, 0x1C, 0xA6, 0x9E, 0x85, 0x50, 0xA4, 0x91, 0x4D, 0xE3, 0xFA, 0xA6, 0xA1, 0x8C, 0x07, 0x2C, 0xC0, 0x12, 0x08, 0x07, 0x3A, 0x92, 0x8D, 0x5D, 0x66, 0xD5, 0x9E, 0xF7, 0x9E, 0x49, 0xA4, 0x29, 0xC4, 0x1A, 0x6B, 0x26, 0x95, 0x71, 0xD5, 0x7E, 0xDB, 0x25, 0xFB, 0xDB, 0x18, 0x38, 0x42, 0x56, 0x08, 0xB4, 0x13, 0xCD, 0x61, 0x6A, 0x5F, 0x6D, 0xB5, 0xB6, 0x07, 0x1A, 0xF9, 0x9B, 0xEA
PolicyB_{SM3_256}	0x16, 0x78, 0x60, 0xA3, 0x5F, 0x2C, 0x5C, 0x35, 0x67, 0xF9, 0xC9, 0x27, 0xAC, 0x56, 0xC0, 0x32, 0xF3, 0xB3, 0xA6, 0x46, 0x2F, 0x8D, 0x03, 0x79, 0x98, 0xE7, 0xA1, 0x0F, 0x77, 0xFA, 0x45, 0x4A

5.4.6 EK Templates H-x

5.4.6.1 Template RSA (Storage)

Table 9 defines the Default EK Template for RSA storage keys.

Parameter differences between different templates are highlighted with **bold** text.

Table 9 Default EK Template (TPMT_PUBLIC): RSA (Storage)

Parameter	Type	RSA 2k (H-1)	RSA 3k (H-6)	RSA 4k (H-7)
type	TPMI_ALG_PUBLIC	TPM_ALG_RSA	TPM_ALG_RSA	TPM_ALG_RSA
nameAlg	TPMI_ALG_HASH	TPM_ALG_SHA256	TPM_ALG_SHA384	TPM_ALG_SHA384
objectAttributes	TPMA_OBJECT	attributes-storage (see Table 7)	attributes-storage (see Table 7)	attributes-storage (see Table 7)
authPolicy	TPM2B_DIGEST			
size	UINT16	32	48	48
buffer	BYTE	PolicyB_{SHA256} (see Table 8)	PolicyB_{SHA384} (see Table 8)	PolicyB_{SHA384} (see Table 8)
parameters	TPMS_RSA_PARMS			
symmetric->algorithm	TPMI_ALG_SYM_OBJECT	TPM_ALG_AES	TPM_ALG_AES	TPM_ALG_AES
symmetric->keyBits	TPMI_AES_KEY_BITS	128	256	256
symmetric->mode	TPMI_ALG_SYM_MODE	TPM_ALG_CFB	TPM_ALG_CFB	TPM_ALG_CFB
scheme->scheme	TPMI_ALG_ASYM_SCHEME	TPM_ALG_NULL	TPM_ALG_NULL	TPM_ALG_NULL
keyBits	TPMI_RSA_KEY_BITS	2048	3072	4096
exponent	UINT32	0	0	0
unique	TPM2B_PUBLIC_KEY_RSA			
size	UINT16	0	0	0
buffer	BYTE	Empty	Empty	Empty

Note: The selection of SHA384 and AES256 to be used with RSA 3k and RSA 4k follows CNSA-Suite recommendations [22].

5.4.6.2 Template RSA (Signing)

Table 10 defines the Default EK Templates for RSA signing keys using either RSA-SSA or RSA-PSS.

Parameter differences between different templates are highlighted with **bold** text.

Table 10 Default EK Template (TPMT_PUBLIC): RSA (Signing)

Parameter	Type	RSA 2k (H-8 or H-9)	RSA 3k (H-14 or H-15)	RSA 4k (H-16 or H-17)
type	TPMI_ALG_PUBLIC	TPM_ALG_RSA	TPM_ALG_RSA	TPM_ALG_RSA
nameAlg	TPMI_ALG_HASH	TPM_ALG_SHA256	TPM_ALG_SHA384	TPM_ALG_SHA384
objectAttributes	TPMA_OBJECT	attributes-signing (see Table 7)	attributes-signing (see Table 7)	attributes-signing (see Table 7)
authPolicy	TPM2B_DIGEST			
size	UINT16	32	48	48
buffer	BYTE	PolicyB_{SHA256} (see Table 8)	PolicyB_{SHA384} (see Table 8)	PolicyB_{SHA384} (see Table 8)
parameters	TPMS_RSA_PARMS			
symmetric->algorithm	TPMI_ALG_SYM_OBJECT	TPM_ALG_NULL	TPM_ALG_NULL	TPM_ALG_NULL
scheme->scheme	TPMI_ALG_ASYM_SCHEME	For RSA-SSA (H-8): TPM_ALG_RSASSA For RSA-PSS (H-9): TPM_ALG_RSAPSS	For RSA-SSA (H-14): TPM_ALG_RSASSA For RSA-PSS (H-15): TPM_ALG_RSAPSS	For RSA-SSA (H-16): TPM_ALG_RSASSA For RSA-PSS (H-17): TPM_ALG_RSAPSS
scheme->details	TPMI_ALG_HASH	TPM_ALG_SHA256	TPM_ALG_SHA384	TPM_ALG_SHA384
keyBits	TPMI_RSA_KEY_BITS	2048	3072	4096
exponent	UINT32	0	0	0
unique	TPM2B_PUBLIC_KEY_RSA			
size	UINT16	0	0	0
buffer	BYTE	Empty	Empty	Empty

Note: The selection of SHA384 and AES256 to be used with RSA 3k and RSA 4k follows CNSA-Suite recommendations [22].

5.4.6.3 Template ECC (Storage)

Table 11 defines the Default EK Templates for ECC storage keys, either regular (non-Firmware-limited) or Firmware-limited.

Parameter differences between different templates are highlighted with **bold** text.

Table 11 Default EK Template (TPMT_PUBLIC): ECC (Storage)

Parameter	Type	NIST P256 (H-2 or H-18)	NIST P384 (H-3 or H-19)	NIST P521 (H-4 or H-20)	SM2 P256 (H-5 or H-21)
type	TPMI_ALG_PUBLIC	TPM_ALG_ECC	TPM_ALG_ECC	TPM_ALG_ECC	TPM_ALG_ECC
nameAlg	TPMI_ALG_HASH	TPM_ALG_SHA256	TPM_ALG_SHA384	TPM_ALG_SHA512	TPM_ALG_SM3_256
objectAttributes	TPMA_OBJECT	For non-FW-limited (H-2): attributes-storage For FW-limited (H-18): attributes-fwl-storage (see Table 7)	For non-FW-limited (H-3): attributes-storage For FW-limited (H-19): attributes-fwl-storage (see Table 7)	For non-FW-limited (H-4): attributes-storage For FW-limited (H-20): attributes-fwl-storage (see Table 7)	For non-FW-limited (H-5): attributes-storage For FW-limited (H-21): attributes-fwl-storage (see Table 7)
authPolicy	TPM2B_DIGEST				
size	UINT16	32	48	64	32
buffer	BYTE	PolicyB_{SHA256} (see Table 8)	PolicyB_{SHA384} (see Table 8)	PolicyB_{SHA512} (see Table 8)	PolicyB_{SM3_256} (see Table 8)
parameters	TPMS_ECC_PARMS				
symmetric->algorithm	TPMI_ALG_SYM_OBJECT	TPM_ALG_AES	TPM_ALG_AES	TPM_ALG_AES	TPM_ALG_SM4
symmetric->keyBits	TPMI_AES_KEY_BITS	128	256	256	128
symmetric->mode	TPMI_ALG_SYM_MODE	TPM_ALG_CFB	TPM_ALG_CFB	TPM_ALG_CFB	TPM_ALG_CFB
scheme->scheme	TPMI_ALG_ECC_SCHEME	TPM_ALG_NULL	TPM_ALG_NULL	TPM_ALG_NULL	TPM_ALG_NULL
curveID	TPMI_ECC_CURVE	TPM_ECC_NIST_P256	TPM_ECC_NIST_P384	TPM_ECC_NIST_P521	TPM_ECC_SM2_P256
kdf->scheme	TPMI_ALG_KDF	TPM_ALG_NULL	TPM_ALG_NULL	TPM_ALG_NULL	TPM_ALG_NULL
unique	TPMS_ECC_POINT				
x->size	UINT16	0	0	0	0
x->buffer	BYTE	Empty	Empty	Empty	Empty
y->size	UINT16	0	0	0	0
y->buffer	BYTE	Empty	Empty	Empty	Empty

Note: For NIST P384, AES 256 bit is used instead of AES 192 bit because a platform specific profile might define AES 192 as optional algorithm.

5.4.6.4 Template ECC (Signing)

Table 12 defines the Default EK Templates for ECC signing keys, either regular (non-Firmware-limited) or Firmware-limited.

Parameter differences between different templates are highlighted with **bold** text.

Table 12 Default EK Template (TPMT_PUBLIC): ECC (Signing)

Parameter	Type	NIST P256 (H-10 or H-22)	NIST P384 (H-11 or H23)	NIST P521 (H-12 or H-24)	SM2 P256 (H-13 or H-25)
type	TPMI_ALG_PUBLIC	TPM_ALG_ECC	TPM_ALG_ECC	TPM_ALG_ECC	TPM_ALG_ECC
nameAlg	TPMI_ALG_HASH	TPM_ALG_SHA256	TPM_ALG_SHA384	TPM_ALG_SHA512	TPM_ALG_SM3_256
objectAttributes	TPMA_OBJECT	For non-FW-limited (H-10): attributes-signing For FW-limited (H-22): attributes-fwl-signing (see Table 7)	For non-FW-limited (H-11): attributes-signing For FW-limited (H-23): attributes-fwl-signing (see Table 7)	For non-FW-limited (H-12): attributes-signing For FW-limited (H-24): attributes-fwl-signing (see Table 7)	For non-FW-limited (H-13): attributes-signing For FW-limited (H-25): attributes-fwl-signing (see Table 7)
authPolicy	TPM2B_DIGEST				
size	UINT16	32	48	64	32
buffer	BYTE	PolicyB_{SHA256} (see Table 8)	PolicyB_{SHA384} (see Table 8)	PolicyB_{SHA512} (see Table 8)	PolicyB_{SM3_256} (see Table 8)
parameters	TPMS_ECC_PARMS				
symmetric->algorithm	TPMI_ALG_SYM_OBJECT	TPM_ALG_NULL	TPM_ALG_NULL	TPM_ALG_NULL	TPM_ALG_NULL
scheme->scheme	TPMI_ALG_ECC_SCHEME	TPM_ALG_ECDSA	TPM_ALG_ECDSA	TPM_ALG_ECDSA	TPM_ALG_SM2
scheme->details	TPMI_ALG_HASH	TPM_ALG_SHA256	TPM_ALG_SHA384	TPM_ALG_SHA512	TPM_ALG_SM3_256
curveID	TPMI_ECC_CURVE	TPM_ECC_NIST_P256	TPM_ECC_NIST_P384	TPM_ECC_NIST_P521	TPM_ECC_SM2_P256
kdf->scheme	TPMI_ALG_KDF	TPM_ALG_NULL	TPM_ALG_NULL	TPM_ALG_NULL	TPM_ALG_NULL
unique	TPMS_ECC_POINT				
x->size	UINT16	0	0	0	0
x->buffer	BYTE	Empty	Empty	Empty	Empty
y->size	UINT16	0	0	0	0
y->buffer	BYTE	Empty	Empty	Empty	Empty

Note: For NIST P384, AES 256 bit is used instead of AES 192 bit because a platform specific profile might define AES 192 as optional algorithm.

5.4.6.5 Template ML-KEM (Storage)

Table 13 defines the Default EK Templates for ML-KEM storage keys, either regular (non-Firmware-limited) or Firmware-limited.

Parameter differences between different templates are highlighted with **bold** text.

Table 13 Default EK Template (TPMT_PUBLIC): ML-KEM (Storage)

Parameter	Type	ML-KEM-512 (H-26 or H-29)	ML-KEM-768 (H-27 or H-30)	ML-KEM-1024 (H-28 or H-31)
type	TPMI_ALG_PUBLIC	TPM_ALG_MLKEM	TPM_ALG_MLKEM	TPM_ALG_MLKEM
nameAlg	TPMI_ALG_HASH	TPM_ALG_SHA256	TPM_ALG_SHA384	TPM_ALG_SHA512
objectAttributes	TPMA_OBJECT	For non-FW-limited (H-26): attributes-storage For FW-limited (H-29): attributes-fwl-storage (see Table 7)	For non-FW-limited (H-27): attributes-storage For FW-limited (H-30): attributes-fwl-storage (see Table 7)	For non-FW-limited (H-28): attributes-storage For FW-limited (H-31): attributes-fwl-storage (see Table 7)
authPolicy	TPM2B_DIGEST			
size	UINT16	32	48	64
buffer	BYTE	PolicyB_{SHA256} (see Table 8)	PolicyB_{SHA384} (see Table 8)	PolicyB_{SHA512} (see Table 8)
parameters	TPMS_MLKEM_PARMS			
symmetric->algorithm	TPMI_ALG_SYM_OBJECT	TPM_ALG_AES	TPM_ALG_AES	TPM_ALG_AES
symmetric->keyBits	TPMI_AES_KEY_BITS	128	256	256
symmetric->mode	TPMI_ALG_SYM_MODE	TPM_ALG_CFB	TPM_ALG_CFB	TPM_ALG_CFB
parameterSet	TPMI_MLKEM_PARMS	TPM_MLKEM_512	TPM_MLKEM_768	TPM_MLKEM_1024
unique	TPM2B_PUBLIC_KEY_MLKEM			
size	UINT16	0	0	0
buffer	BYTE	Empty	Empty	Empty

5.4.6.6 Template ML-DSA (Signing)

Table 14 defines the Default EK Templates for ML-DSA signing keys, either regular (non-Firmware-limited) or Firmware-limited.

Parameter differences between different templates are highlighted with **bold** text.

Table 14 Default EK Template (TPMT_PUBLIC): ML-DSA (Signing)

Parameter	Type	ML-DSA-44 (H-32 or H-35)	ML-DSA-65 (H-33 or H-36)	ML-DSA-87 (H-34 or H-37)
type	TPMI_ALG_PUBLIC	TPM_ALG_MLDSA	TPM_ALG_MLDSA	TPM_ALG_MLDSA
nameAlg	TPMI_ALG_HASH	TPM_ALG_SHA256	TPM_ALG_SHA384	TPM_ALG_SHA512
objectAttributes	TPMA_OBJECT	For non-FW-limited (H-32): attributes-signing For FW-limited (H-35): attributes-fwl-signing (see Table 7)	For non-FW-limited (H-33): attributes-signing For FW-limited (H-36): attributes-fwl-signing (see Table 7)	For non-FW-limited (H-34): attributes-signing For FW-limited (H-37): attributes-fwl-signing (see Table 7)
authPolicy	TPM2B_DIGEST			
size	UINT16	32	48	64
buffer	BYTE	PolicyB_{SHA256} (see Table 8)	PolicyB_{SHA384} (see Table 8)	PolicyB_{SHA512} (see Table 8)
parameters	TPMS_MLDSA_PARMS			
parameterSet	TPMI_MLDSA_PARMS	TPM_MLDSA_44	TPM_MLDSA_65	TPM_MLDSA_87
allowExternalMu	BOOL	0	0	0
unique	TPM2B_PUBLIC_KEY_MLDSA			
size	UINT16	0	0	0
buffer	BYTE	Empty	Empty	Empty

6 Certificate Fields (algorithm-specific)

6.1 Signature Algorithm

The signature algorithm depends on the algorithm of the CA key used to sign the EK Certificate. The security strength of the signing algorithm SHOULD be equivalent to the security strength of the signing key. The security strength of the CA key used to sign the EK Certificate SHALL have an equal or higher security strength than the EK. An EK Certificate for an RSA EK MAY be signed using ECDSA with an ECC NIST P256, 384, or 521 CA Key.

6.1.1 RSA

When using an RSA CA key, the EK Certificate SHOULD be signed using the algorithms appropriate to the key size. The AlgorithmIdentifier parameters field MUST be the ASN.1 type NULL.

6.1.1.1 RSA 2k CA Key

For an RSA 2k CA key, the algorithm SHOULD be sha256WithRSAEncryption as defined in RFC 5754 [13].

```
sha256WithRSAEncryption OBJECT IDENTIFIER ::= {  
    iso(1) member-body(2) us(840) rsadsi(113549) pkcs(1) pkcs-1(1) 11 }
```

6.1.1.2 RSA 3k and 4k CA Key

For an RSA 3k or 4k CA key, the algorithm SHOULD be sha384WithRSAEncryption as defined in RFC 5754 [13].

```
sha384WithRSAEncryption OBJECT IDENTIFIER ::= {  
    iso(1) member-body(2) us(840) rsadsi(113549) pkcs(1) pkcs-1(1) 12 }
```

Note: 1 The use of SHA384 with RSA 3k and 4k follows CNSA-Suite recommendations [22].

Note: 2 The recommendation for use of sha384WithRSAEncryption for RSA 3k and 4k CA key was added in version 2.3 of this specification.

6.1.2 ECC

When using an ECC CA key, the EK Certificate SHOULD be signed using the algorithms appropriate to the curve size. The AlgorithmIdentifier parameters field MUST be absent.

6.1.2.1 NIST P256 CA Key

For an ECC NIST P256 CA key, the algorithm SHOULD be ecdsa-with-SHA256 as defined in RFC 5754 [13].

```
ecdsa-with-SHA256 OBJECT IDENTIFIER ::= { iso(1) member-body(2)  
    us(840) ansi-X9-62(10045) signatures(4) ecdsa-with-SHA2(3) 2 }
```

6.1.2.2 NIST P384 CA Key

For an ECC NIST P384 CA key, the algorithm SHOULD be ecdsa-with-SHA384 as defined in RFC 5754 [13].

```
ecdsa-with-SHA384 OBJECT IDENTIFIER ::= {  
    iso(1) member-body(2) us(840) ansi-X9-62(10045) signatures(4) ecdsa-with-SHA2(3) 3 }
```

6.1.2.3 NIST P521 CA Key

For an ECC NIST P521 CA key, the algorithm SHOULD be ecdsa-with-SHA512 as defined in RFC 5754 [13].

```
ecdsa-with-SHA512 OBJECT IDENTIFIER ::= {  
    iso(1) member-body(2) us(840) ansi-X9-62(10045) signatures(4) ecdsa-with-SHA2(3) 4 }
```

6.1.2.4 SM2 P256 CA Key

When using an ECC SM2 key, the EK Certificate SHOULD be signed using the algorithm SM3WithSM2Encryption which has the OID value defined in GM/T 0006-2012 Cryptographic

Application Identifier Criterion Specification [20] as shown below. The AlgorithmIdentifier parameters field MUST be absent.

```
SM3WithSM2Encryption OBJECT IDENTIFIER ::= { iso(1) member-body(2) cn(156)
ccstc(10197) cryptographic-algorithm (1) 501 }
```

6.1.3 ML-DSA

For an ML-DSA CA key, the algorithm SHOULD be one of the ML-DSA algorithm identifiers defined in RFC TBD ([draft-ietf-lamps-dilithium-certificates](#))[25].

6.2 Subject Public Key Info

6.2.1 RSA

For an RSA public key the algorithm rsaEncryption that has the OID value defined in RFC 3279 [14] as shown below MUST be used. The AlgorithmIdentifier parameters field MUST be the ASN.1 type NULL.

```
rsaEncryption OBJECT IDENTIFIER ::= {
iso(1) member-body(2) us(840) rsadsi(113549) pkcs(1) pkcs-1(1) 1 }
```

The RSA public key MUST be encoded using the ASN.1 type RSAPublicKey as defined in RFC 3279 [14].

```
RSAPublicKey ::= SEQUENCE {
    modulus          INTEGER,      -- n
    publicExponent   INTEGER }    -- e
```

6.2.2 ECC

For an ECC public key the algorithm id-ecPublicKey which has the OID value defined in RFC 5480 [15] as shown below MUST be used. The ECParameters field is required, the nameCurve field SHOULD contain the OID of the respective curve (see below).

```
id-ecPublicKey OBJECT IDENTIFIER ::= {
iso(1) member-body(2) us(840) ansi-X9-62(10045) keyType(2) 1 }
```

```
ECParameters ::= CHOICE {
    namedCurve          OBJECT IDENTIFIER
    -- implicitCurve    NULL
    -- specifiedCurve   SpecifiedECDomain
}
```

The ECC public key MUST be encoded as an ECC Point. The uncompressed format SHOULD be used.

```
ECPoint ::= OCTET STRING
```

The namedCurve field in ECParameters of the Subject Public Key Info depends on the ECC curve.

6.2.2.1 NIST P256

For NIST P256, the namedCurve field MUST contain the OID defined in RFC 5480 [15]:

```
secp256r1 OBJECT IDENTIFIER ::= {
iso(1) member-body(2) us(840) ansi-X9-62(10045) curves(3) prime(1) 7 }
```

6.2.2.2 NIST P384

For NIST P384, the namedCurve field MUST contain the OID defined in RFC 5480 [15]:

```
secp384r1 OBJECT IDENTIFIER ::= {
iso(1) identified-organization(3) certicom(132) curve(0) 34 }
```

6.2.2.3 NIST P521

For NIST P521, the namedCurve field MUST contain the OID defined in RFC 5480 [15]:

```
secp521r1 OBJECT IDENTIFIER ::= {
    iso(1) identified-organization(3) certicom(132) curve(0) 35 }
```

6.2.2.4 SM2 P256

For SM2 P256, the namedCurve field MUST contain the OID defined in GM/T 0006-2012 Cryptographic Application Identifier Criterion Specification [20]:

```
SM2EllipticCurveCryptography OBJECT IDENTIFIER ::= {
    iso(1) member-body(2) cn(156) ccsc(10197) cryptographic-algorithm (1) 301 }
```

6.2.3 ML-KEM

For an ML-KEM public key, the public key MUST be encoded using one of the ML-KEM public key types defined in RFC TBD ([draft-ietf-lamps-kyber-certificates](#)) [26].

```
nistAlgorithms OBJECT IDENTIFIER ::= { joint-iso-ccitt(2)
    country(16) us(840) organization(1) gov(101) csor(3)
    nistAlgorithm(4) }

kems OBJECT IDENTIFIER ::= { nistAlgorithms 4 }

id-alg-ml-kem-512 OBJECT IDENTIFIER ::= { kems 1 }

id-alg-ml-kem-768 OBJECT IDENTIFIER ::= { kems 2 }

id-alg-ml-kem-1024 OBJECT IDENTIFIER ::= { kems 3 }
```

6.2.4 ML-DSA

For an ML-DSA public key, the public key MUST be encoded using one of the ML-DSA public key types defined in RFC TBD ([draft-ietf-lamps-dilithium-certificates](#)) [25].

7 References

For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

- [1] TPM 2.0 Library Specification:
http://www.trustedcomputinggroup.org/resources/tpm_library_specification
- [2] Registry of Reserved TPM 2.0 Handles and Localities:
<http://www.trustedcomputinggroup.org/resources/registry>
- [3] Vendor ID Registry:
http://www.trustedcomputinggroup.org/resources/vendor_id_registry
- [4] TCG Credential Profile Version 1.0, Version 1.1 and Version 1.2:
http://www.trustedcomputinggroup.org/resources/infrastructure_work_group_tcg_credential_profiles_specification
- [5] TCG EK Credential Profile for TPM 2.0, Version 2.0, Revision 14:
<https://trustedcomputinggroup.org/tcg-ek-credential-profile-tpm-family-2-0>
- [6] PC Client Specific Platform TPM Profile for TPM 2.0:
http://www.trustedcomputinggroup.org/developers/pc_client
- [7] CMC Profile for EK/Platform Certificate Enrollment for TPM 1.2:
<http://www.trustedcomputinggroup.org/developers/infrastructure>
- [8] TPM Keys for Platform Identity for TPM 1.2:
<http://www.trustedcomputinggroup.org/developers/infrastructure>
- [9] TPM 1.2 Main Specification:
http://www.trustedcomputinggroup.org/resources/tpm_main_specification
- [10] IEEE Standard for Local and metropolitan area networks, Secure Device Identity, 2009
- [11] NIST Special Publication 800-57, Recommendation for Key Management – Part 1: General
- [12] Internet X.509 Public Key Infrastructure Certificate and Certificate Revocation List (CRL) Profile, RFC 5280, <http://www.ietf.org/rfc/rfc5280.txt>
- [13] Using SHA2 Algorithms with Cryptographic Message Syntax, RFC 5754,
<http://www.ietf.org/rfc/rfc5754.txt>
- [14] Algorithms and Identifiers for the Internet X.509 Public Key Infrastructure Certificate and Certificate Revocation List (CRL) Profile, RFC 3279, <http://www.ietf.org/rfc/rfc3279.txt>
- [15] Elliptic Curve Cryptography Subject Public Key Information, RFC 5480,
<http://www.ietf.org/rfc/rfc5480.txt>
- [16] Using Cryptographic Message Syntax (CMS) to Protect Firmware Packages, RFC 4108,
<http://www.ietf.org/rfc/rfc4108.txt>
- [17] Key words for use in RFCs to Indicate Requirement Levels, RFC 2119,
<http://www.ietf.org/rfc/rfc2119.txt>
- [18] Hypertext Markup Language – 2.0, RFC 1866, <http://www.ietf.org/rfc/rfc1866.txt>
- [19] ITU-T X.690: Information technology – ASN.1 encoding rules: Specification of Basic Encoding Rules (BER), Canonical Encoding Rules (CER) and Distinguished Encoding Rules (DER)
- [20] Cryptographic Application Identifier Criterion Specification, GM/T 0006-2012,
http://www.cssn.net.cn/pagesnew/search/standard_detail.jsp?a001=NjM0ODU5Mg==
- [21] TCG Glossary: <https://trustedcomputinggroup.org/glossary>
- [22] CNSA-Suite-and-Quantum-Computing-FAQ.pdf, <https://www.nsa.gov/>

- [23]** DICE Attestation Architecture: <https://trustedcomputinggroup.org/resource/dice-attestation-architecture/>
- [24]** Automotive Thin Profile: <https://trustedcomputinggroup.org/resource/tcg-tpm-2-0-library-profile-for-automotive-thin/>
- [25]** Internet X.509 Public Key Infrastructure - Algorithm Identifiers for the Module-Lattice-Based Digital Signature Algorithm (ML-DSA) RFC TBD, <https://datatracker.ietf.org/doc/draft-ietf-lamps-dilithium-certificates/>
- [26]** Internet X.509 Public Key Infrastructure - Algorithm Identifiers for the Module-Lattice-Based Key-Encapsulation Mechanism (ML-KEM) RFC TBD , <https://datatracker.ietf.org/doc/draft-ietf-lamps-kyber-certificates/>
- [27]** ISO/IEC/ITU-T X.509 version 3 <https://www.itu.int/rec/t-rec-x.509>

A. EK Template Related Definitions

A.1 Policy NV Indices

A.1.1 Introduction

This annex defines the public structure (TPMS_NV_PUBLIC) of NV Indices that may be used to store a policy digest in their data (payload) area. This policy may then be used as authorization policy by an Endorsement Key that was created with one of the default EK Templates of the High Range. Other authorizations options – which do not require such an NV Index - are described in 5.4.

One NV Index is reserved for each hash algorithm: SHA256, SHA384, SHA512 and SM3_256. Figure 1 illustrates which Policy Index (I-1 to I-4) is to be used by which default EK Template of the High Range (H-1 to H-5).

The Policy NV Indices are not *Populated* by the TPM manufacturer. Instead, a Policy NV Index can be defined and undefined by the platform Owner using Owner Authorization. The NV Index is written with Endorsement authorization.

Note: It is not anticipated that all four Policy NV Indices would be *Populated* at the same time in a TPM. Instead, only the Policy Index corresponding to the EK in operation would be *Populated*. For example, Policy Index I-1 for use of EKs created with Template H-1 or H-2.

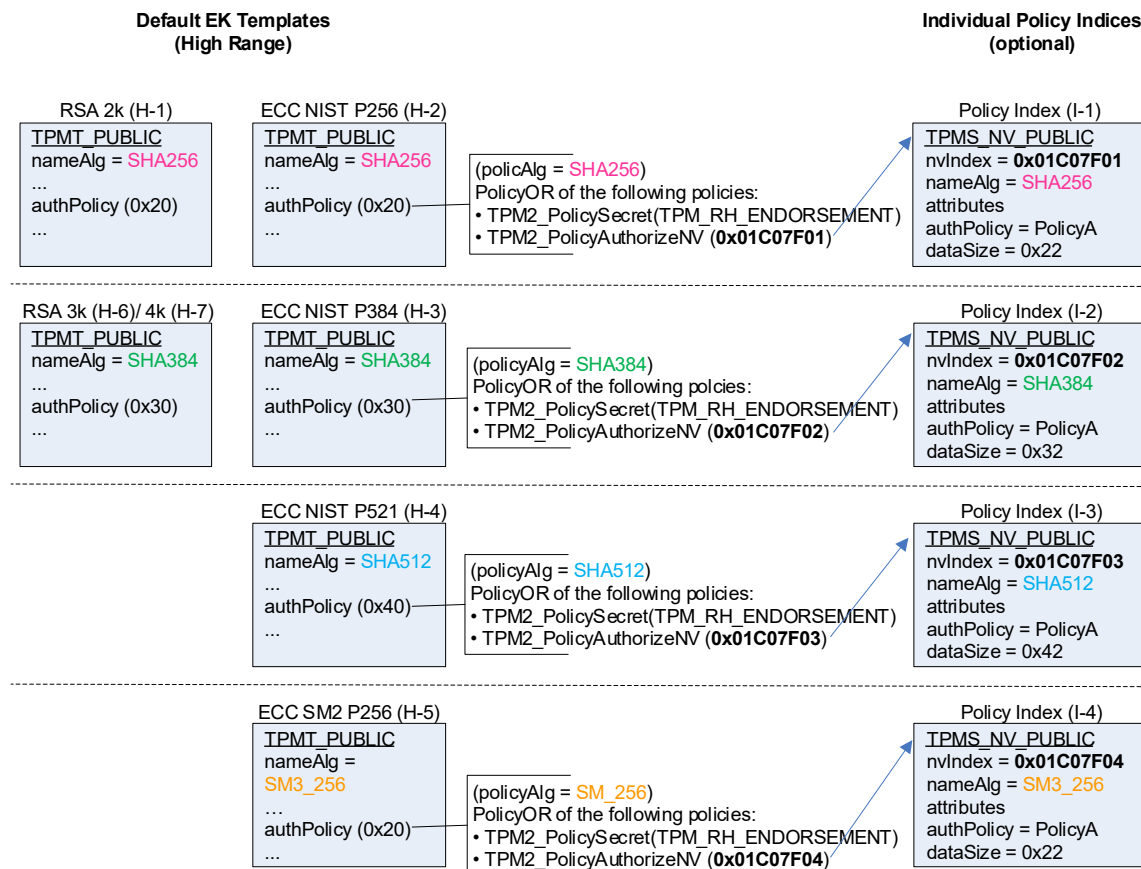


Figure 1: Policy Indices for EK Templates in the High Range

A.1.2 Handle Values

The Policy NV Indices (if *Populated*) use the following handle values:

0x01c07f01	Policy Index I-1 with nameAlg = SHA256 (A.1.3)
0x01c07f02	Policy Index I-2 with nameAlg = SHA384 (A.1.4)
0x01c07f03	Policy Index I-3 with nameAlg = SHA512 (A.1.5)
0x01c07f04	Policy Index I-4 with nameAlg = SM3_256 (A.1.6)

A.1.3 Policy Index I-1: SHA256

Parameter	Type	Content
nvIndex	TPMI_RH_NV_INDEX	0x01C07F01
nameAlg	TPMI_ALG_HASH	TPM_ALG_SHA256 (0x000B)
attributes	TPMA_NV	TPMA_NV_PPWRITE = 0 TPMA_NV_OWNERWRITE = 0 TPMA_NV_AUTHWRITE = 0 TPMA_NV_POLICYWRITE = 1 TPM_NT = 0 TPMA_NV_POLICY_DELETE = 0 TPMA_NV_WRITELOCKED = 0 TPMA_NV_WRITEALL = 1 TPMA_NV_WRITEDEFINE = 0 TPMA_NV_WRITE_STCLEAR = 0 TPMA_NV_GLOBALLOCK = 0 TPMA_NV_PPREAD = 1 TPMA_NV_OWNERREAD = 1 TPMA_NV_AUTHREAD = 1 TPMA_NV_POLICYREAD = 1 TPMA_NV_NO_DA = 1 TPMA_NV_ORDERLY = 0 TPMA_NV_CLEAR_STCLEAR = 0 TPMA_NV_READLOCKED = 0 TPMA_NV_WRITTEN = 1 TPMA_NV_PLATFORMCREATE = 0 TPMA_NV_READ_STCLEAR = 0 (0x220F1008)
authPolicy	TPM2B_DIGEST	
size	UINT16	32 (0x0020)
buffer	BYTE	0x83, 0x71, 0x97, 0x67, 0x44, 0x84, 0xB3, 0xF8, 0x1A, 0x90, 0xCC, 0x8D, 0x46, 0xA5, 0xD7, 0x24, 0xFD, 0x52, 0xD7, 0x6E, 0x06, 0x52, 0x0B, 0x64, 0xF2, 0xA1, 0xDA, 0x1B, 0x33, 0x14, 0x69, 0xAA (PolicyA _{SHA256} , see Table 19)
dataSize	UINT16	34 (0x0022)

Table 15: EK Policy Index (TPMS_NV_PUBLIC) I-1: SHA256

Note: The first two bytes of the Index data area contain a TPM_ALG_ID, followed by the policy value (without size).

A.1.4 Policy Index I-2: SHA384

Parameter	Type	Content
nvIndex	TPMI_RH_NV_INDEX	0x01C07F02
nameAlg	TPMI_ALG_HASH	TPM_ALG_SHA384 (0x000C)
attributes	TPMA_NV	TPMA_NV_PPWRITE = 0 TPMA_NV_OWNERWRITE = 0 TPMA_NV_AUTHWRITE = 0 TPMA_NV_POLICYWRITE = 1 TPM_NT = 0 TPMA_NV_POLICY_DELETE = 0 TPMA_NV_WRITELOCKED = 0 TPMA_NV_WRITEALL = 1 TPMA_NV_WRITEDEFINE = 0 TPMA_NV_WRITE_STCLEAR = 0 TPMA_NV_GLOBALLOCK = 0 TPMA_NV_PPREAD = 1 TPMA_NV_OWNERREAD = 1 TPMA_NV_AUTHREAD = 1 TPMA_NV_POLICYREAD = 1 TPMA_NV_NO_DA = 1 TPMA_NV_ORDERLY = 0 TPMA_NV_CLEAR_STCLEAR = 0 TPMA_NV_READLOCKED = 0 TPMA_NV_WRITTEN = 1 TPMA_NV_PLATFORMCREATE = 0 TPMA_NV_READ_STCLEAR = 0 (0x220F1008)
authPolicy	TPM2B_DIGEST	
size	UINT16	48 (0x0030)
buffer	BYTE	0x8B, 0xBF, 0x22, 0x66, 0x53, 0x7C, 0x17, 0x1C, 0xB5, 0x6E, 0x40, 0x3C, 0x4D, 0xC1, 0xD4, 0xB6, 0x4F, 0x43, 0x26, 0x11, 0xDC, 0x38, 0x6E, 0x6F, 0x53, 0x20, 0x50, 0xC3, 0x27, 0x8C, 0x93, 0x0E, 0x14, 0x3E, 0x8B, 0xB1, 0x13, 0x38, 0x24, 0xCC, 0xB4, 0x31, 0x05, 0x38, 0x71, 0xC6, 0xDB, 0x53 (PolicyASHA384, see Table 19)
dataSize	UINT16	50 (0x0032)

Table 16: EK Policy Index (TPMS_NV_PUBLIC) I-2: SHA384

Note: The first two bytes of the Index data area contain a TPM_ALG_ID, followed by the policy value (without size).

A.1.5 Policy Index I-3: SHA512

Parameter	Type	Content
nvIndex	TPMI_RH_NV_INDEX	0x01C07F03
nameAlg	TPMI_ALG_HASH	TPM_ALG_SHA512 (0x000D)
attributes	TPMA_NV	TPMA_NV_PPWRITE = 0 TPMA_NV_OWNERWRITE = 0 TPMA_NV_AUTHWRITE = 0 TPMA_NV_POLICYWRITE = 1 TPM_NT = 0 TPMA_NV_POLICY_DELETE = 0 TPMA_NV_WRITELOCKED = 0 TPMA_NV_WRITEALL = 1 TPMA_NV_WRITEDEFINE = 0 TPMA_NV_WRITE_STCLEAR = 0 TPMA_NV_GLOBALLOCK = 0 TPMA_NV_PPREAD = 1 TPMA_NV_OWNERREAD = 1 TPMA_NV_AUTHREAD = 1 TPMA_NV_POLICYREAD = 1 TPMA_NV_NO_DA = 1 TPMA_NV_ORDERLY = 0 TPMA_NV_CLEAR_STCLEAR = 0 TPMA_NV_READLOCKED = 0 TPMA_NV_WRITTEN = 1 TPMA_NV_PLATFORMCREATE = 0 TPMA_NV_READ_STCLEAR = 0 (0x220F1008)
authPolicy	TPM2B_DIGEST	
size	UINT16	64 (0x0040)
buffer	BYTE	0x1E, 0x3B, 0x76, 0x50, 0x2C, 0x8A, 0x14, 0x25, 0xAA, 0x0B, 0x7B, 0x3F, 0xC6, 0x46, 0xA1, 0xB0, 0xFA, 0xE0, 0x63, 0xB0, 0x3B, 0x53, 0x68, 0xF9, 0xC4, 0xCD, 0xDE, 0xCA, 0xFF, 0x08, 0x91, 0xDD, 0x68, 0x2B, 0xAC, 0x1A, 0x85, 0xD4, 0xD8, 0x32, 0xB7, 0x81, 0xEA, 0x45, 0x19, 0x15, 0xDE, 0x5F, 0xC5, 0xBF, 0x0D, 0xC4, 0xA1, 0x91, 0x7C, 0xD4, 0x2F, 0xA0, 0x41, 0xE3, 0xF9, 0x98, 0xE0, 0xEE (PolicyA _{SHA512} , see Table 19)
dataSize	UINT16	66 (0x0042)

Table 17: EK Policy Index (TPMS_NV_PUBLIC) I-3: SHA512

Note: The first two bytes of the Index data area contain a TPM_ALG_ID, followed by the policy value (without size).

A.1.6 Policy Index I-4: SM3_256

Parameter	Type	Content
nvIndex	TPMI_RH_NV_INDEX	0x01C07F04
nameAlg	TPMI_ALG_HASH	TPM_ALG_SM3_256 (0x0012)
attributes	TPMA_NV	TPMA_NV_PPWRITE = 0 TPMA_NV_OWNERWRITE = 0 TPMA_NV_AUTHWRITE = 0 TPMA_NV_POLICYWRITE = 1 TPM_NT = 0 TPMA_NV_POLICY_DELETE = 0 TPMA_NV_WRITELOCKED = 0 TPMA_NV_WRITEALL = 1 TPMA_NV_WRITEDEFINE = 0 TPMA_NV_WRITE_STCLEAR = 0 TPMA_NV_GLOBALLOCK = 0 TPMA_NV_PPREAD = 1 TPMA_NV_OWNERREAD = 1 TPMA_NV_AUTHREAD = 1 TPMA_NV_POLICYREAD = 1 TPMA_NV_NO_DA = 1 TPMA_NV_ORDERLY = 0 TPMA_NV_CLEAR_STCLEAR = 0 TPMA_NV_READLOCKED = 0 TPMA_NV_WRITTEN = 1 TPMA_NV_PLATFORMCREATE = 0 TPMA_NV_READ_STCLEAR = 0 (0x220F1008)
authPolicy	TPM2B_DIGEST	
size	UINT16	32 (0x0020)
buffer	BYTE	0xC6, 0x7F, 0x7D, 0x35, 0xF6, 0x6F, 0x3B, 0xEC, 0x13, 0xC8, 0x9F, 0xE8, 0x98, 0x92, 0x1C, 0x65, 0x1B, 0x0C, 0xB5, 0xA3, 0x8A, 0x92, 0x69, 0x0A, 0x62, 0xA4, 0x3C, 0x00, 0x12, 0xE4, 0xFB, 0x8B (PolicyA _{SM2_256} , see Table 19)
dataSize	UINT16	34 (0x0022)

Table 18: EK Policy Index (TPMS_NV_PUBLIC) I-4: SM3_256

Note: The first two bytes of the Index data area contain a TPM_ALG_ID, followed by the policy value (without size).

A.2 Policy Computation

A.2.1 Introduction

This annex documents how the different policy values used in the default EK Templates (in sections 5.3 and 5.4) and the Policy NV Indices (in A.1) were computed. The equations in this section are copied from the TPM 2.0 Library Specification Part 3 [1]. If there are any inconsistencies between the equations below and the equations defined in the TPM 2.0 Library Specification, the definitions in the Library Specification take precedence.

A.2.2 Computing PolicyA

TPM2_PolicySecret() uses the PolicyUpdate function:

PolicyUpdate(TPM_CC_PolicySecret, authObject→Name, policyRef)

This is equivalent to:

$\text{policyDigest}_{\text{new}} := H_{\text{policyAlg}}(\text{policyDigest}_{\text{old}} \parallel \text{TPM_CC_PolicySecret} \parallel \text{authObject} \rightarrow \text{Name})$
 $\text{policyDigest}_{\text{new}+1} := H_{\text{policyAlg}}(\text{policyDigest}_{\text{new}} \parallel \text{policyRef.buffer})$

With:

policyAlg = SHA256, or SHA384, or SHA512, or SM3_256
policyDigest_{old} = 0x0...0 (32, or 48, or 64 bytes)
TPM_CC_PolicySecret = 0x00000151
authObject→Name is TPM_RH_ENDORSEMENT (=0x4000000B)
policyRef.buffer = not available (policyRef is an Empty Buffer)

The policy digest is calculated as follows:

$\text{policyDigest}_{\text{new}} := H_{\text{policyAlg}}(0x0...0 \parallel 0x00000151 \parallel 0x4000000B)$
 $\text{policyDigest}_{\text{new}+1} := H_{\text{policyAlg}}(\text{policyDigest}_{\text{new}})$

The following table contains the computed *PolicyA* values for the different hash algorithms.

<i>PolicyA</i>	Value (hex)
<i>PolicyA</i> _{SHA256} $H_{\text{SHA256}}(H_{\text{SHA256}}(0x0...0 \parallel 0x00000151 \parallel 0x4000000B))$	837197674484b3f81a90cc8d46a5d724 fd52d76e06520b64f2a1da1b331469aa
<i>PolicyA</i> _{SHA384} $H_{\text{SHA384}}(H_{\text{SHA384}}(0x0...0 \parallel 0x00000151 \parallel 0x4000000B))$	8bbf2266537c171cb56e403c4dc1d4b6 4f432611dc386e6f532050c3278c930e 143e8bb1133824ccb431053871c6db53
<i>PolicyA</i> _{SHA512} $H_{\text{SHA512}}(H_{\text{SHA512}}(0x0...0 \parallel 0x00000151 \parallel 0x4000000B))$	1e3b76502c8a1425aa0b7b3fc646a1b0 fae063b03b5368f9c4cddecaff0891dd 682bac1a85d4d832b781ea451915de5f c5bf0dc4a1917cd42fa041e3f998e0ee
<i>PolicyA</i> _{SM3_256} $H_{\text{SM3_256}}(H_{\text{SM3_256}}(0x0...0 \parallel 0x00000151 \parallel 0x4000000B))$	c67f7d35f66f3bec13c89fe898921c65 1b0cb5a38a92690a62a43c0012e4fb8b

Table 19: PolicyA values

A.2.3 Computing Policy Index Names

The NV Index Name is computed as defined in Part 1:

Name := nameAlg || H_{nameAlg} (handle→nvPublicArea):

Where

handle→nvPublicArea is the public area of the NV Index (TPMS_NV_PUBLIC) and
TPMS_NV_PUBLIC = nvIndex || nameAlg || attributes || authPolicy || dataSize

With:

nvIndex = 0x01C07F01, 0x01C07F02, 0x01C07F03, or 0x01C07F04
nameAlg = TPM_ALG_SHA256/ SHA384/ SHA512, or TPM_ALG_SM2
attributes = 0x220F1008
authPolicy = PolicyA (including size), see Table 19
dataSize = 0x0022, 0x0032, or 0x0042

The name for the Policy Indices I-1 to I-4 (see A.1.3 to A.1.6) is computed as shown in Table 20.

Policy Index Names	Value (hex)
I-1 Name 0x000B H _{SHA256} (0x01C07F01 0x000B 0x220F1008 0x0020 <i>PolicyA</i> _{SHA256} 0x0022)	000b0c9d717e9c3fe69fda41769450bb 145957f8b3610e084dbf65591a5d11ec d83f
I-2 Name 0x000C H _{SHA384} (0x01C07F02 0x000C 0x220F1008 0x0030 <i>PolicyA</i> _{SHA384} 0x0032)	000cdb62fca346612c976732ff4e8621 fb4e858be82586486504f7d02e621f8d 7d61ae32cfc60c4d120609ed6768afcf 090c
I-3 Name 0x000D H _{SHA512} (0x01C07F03 0x000D 0x220F1008 0x0040 <i>PolicyA</i> _{SHA512} 0x0042)	000d1c47c0bbcbd3cf7d7cae6987d319 37c171015dde3b7f0d3c869bca1f7e8a 223b9acfadb49b7c9cf14d450f41e932 7de34d9291eece2c58ab1dc10e9059cc e560
I-4 Name 0x0012 H _{SM3_256} (0x01C07F04 0x0012 0x220F1008 0x0020 <i>PolicyA</i> _{SM3_256} 0x0022)	001298c4652e788dd7ddcccc353a5ea1 a0e0b5efd2e7af1afb09cae8d9453c5f 1152

Table 20: Policy Index Names

A.2.4 Computing PolicyC

The policy digest for TPM2_PolicyAuthorizeNV() is computed as defined in the Library Spec, Part 3:

policyDigest_{new} := H_{policyAlg}(policyDigest_{old} || TPM_CC_PolicyAuthorizeNV || nvIndex→Name)

Where

policyAlg = SHA256, SHA384, SHA512, or SM3_256
policyDigest_{old} = 0x0...0 (32, 48, or 64 bytes)
TPM_CC_PolicyAuthorizeNV = 0x00000192
nvIndex→Name is the Name of the NV Index containing the policy

With the Policy Index Names from Table 20, Policy C is computed as shown in Table 21.

<i>PolicyC</i>	Value (in hex)
<i>PolicyC</i> _{SHA256} H _{SHA256} (0x0...0 0x00000192 Name of Policy Index I-1)	3767e2edd43ff45a3a7e1eaeefcef7864 3dca964632e7aad82c673a30d8633fde
<i>PolicyC</i> _{SHA384} H _{SHA384} (0x0...0 0x00000192 Name of Policy Index I-2)	d6032ce61f2fb3c240eb3cf6a33237ef 2b6a16f4293c22b455e261cffd217ad5 b4947c2d73e63005eed2dc2b3593d165
<i>PolicyC</i> _{SHA512} H _{SHA512} (0x0...0 0x00000192 Name of Policy Index I-3)	589ee1e146544716e8deafe6db247b01 b81e9f9c7dd16b814aa159138749105f ba5388dd1dea702f35240c184933121e 2c61b8f50d3ef91393a49a38c3f73fc8
<i>PolicyC</i> _{SM3_256} H _{SM3_256} (0x0...0 0x00000192 Name of Policy Index I-4)	2d4e81578c3531d9bd1cdd7d02ba298d 5699a3e39fc3551bfefffcf132b49e11d

Table 21: PolicyC values

A.2.5 Computing PolicyB

The policy digest for TPM2_PolicyOR() is computed as defined in the Library Spec, Part 3:

$$\text{policyDigest}_{\text{new}} := \mathbf{H}_{\text{policyAlg}}(\text{policyDigest}_{\text{old}} || \text{TPM_CC_PolicyOR} || \text{digests})$$

Where

$$\text{digests} := \text{pHashList.digests}[1].\text{buffer} || \dots || \text{pHashList.digests}[n].\text{buffer}$$

With

TPM_CC_PolicyOR = 0x00000171
pHashList.digests[1].buffer = PolicyA from Table 19
pHashList.digests[2].buffer = PolicyC from Table 21

PolicyB is computed as shown in Table 22.

<i>PolicyB</i>	Value (in hex)
<i>PolicyB</i> _{SHA256} H _{SHA256} (0x0...0 0x00000171 <i>PolicyA</i> _{SHA256} <i>PolicyC</i> _{SHA256})	ca3d0a99a2b93906f7a3342414efcfb3 a385d44cd1fd459089d19b5071c0b7a0
<i>PolicyB</i> _{SHA384} H _{SHA384} (0x0...0 0x00000171 <i>PolicyA</i> _{SHA384} <i>PolicyC</i> _{SHA384})	b26e7d28d11a50bc53d882bcf5fd3a1a 074148bb35d3b4e4cb1c0ad9bde419ca cb47ba09699646150f9fc000f3f80e12
<i>PolicyB</i> _{SHA512} H _{SHA512} (0x0...0 0x00000171 <i>PolicyA</i> _{SHA512} <i>PolicyC</i> _{SHA512})	b8221ca69e8550a4914de3faa6a18c07 2cc01208073a928d5d66d59ef79e49a4 29c41a6b269571d57edb25fbdb183842 5608b413cd616a5f6db5b6071af99bea
<i>PolicyB</i> _{SM3_256} H _{SM3_256} (0x0...0 0x00000171 <i>PolicyA</i> _{SM3_256} <i>PolicyC</i> _{SM3_256})	167860a35f2c5c3567f9c927ac56c032 f3b3a6462f8d037998e7a10f77fa454a

Table 22: PolicyB values

B. Certificate Examples

B.1 Example 1 (user device TPM, e.g. PC-Client)

This annex provides an example for a standard, user device TPM (e.g. PC-Client) Endorsement Key Certificate. The ASN.1 encoding for the subject alternative name and subject directory attributes extension is provided below. The values used in this example are for illustrative purposes and must be replaced with manufacturer-specific data.

Subject alternative name:

TPMManufacturer = id:54534700 (TCG)

TPMModel = ABCDEF123456 (part number)

TPMVersion = id:00010023 (firmware version)

```
// SEQUENCE
30 49
  // SET
  31 16
    // SEQUENCE
    30 14
      // OBJECT IDENTIFIER tcg-at-tpmManufacturer (2.23.133.2.1)
      06 05 67 81 05 02 01
      // UTF8 STRING id:54434700 (TCG)
      0C 0B 69 64 3A 35 34 34 33 34 37 30 30
    // SET
    31 17
      // SEQUENCE
      30 15
        // OBJECT IDENTIFIER tcg-at-tpmModel (2.23.133.2.2)
        06 05 67 81 05 02 02
        // UTF8 STRING ABCDEF123456
        0C 0C 41 42 43 44 45 46 31 32 33 34 35 36
      // SET
      31 16
        // SEQUENCE
        30 14
          // OBJECT IDENTIFIER tcg-at-tpmVersion (2.23.133.2.3)
          06 05 67 81 05 02 03
          // UTF8 STRING id:00010023
          0C 0B 69 64 3A 30 30 30 31 30 30 32 33
```

Subject directory attributes:

TPMSpecification

Family = id:322E3000 (2.0)

Level = 0

Revision = 99

TPMSecurityAssertions (not included here since optional)

```
// SEQUENCE
30 16
  // OBJECT IDENTIFIER tcg-at-tpmSpecification (2.23.133.2.16)
  06 05 67 81 05 02 10
  // SET
  31 0D
    // SEQUENCE
    30 0B
      // UTF8 STRING (2.0)
      0C 03 32 2E 30
      // INTEGER (0)
      02 01 00
      // INTEGER (99)
      02 01 63
```

The encoding of the extensions above is extracted from the following example certificate. The example certificate provided below is for illustrative purposes only, and all example values must be replaced with manufacturer-specific data. For simplicity some optional configurations (e.g. optional data within an extension) are omitted. The manufacturer's certificate is not required to look exactly the same as this example certificate. For easier testing the certificate is provided in PEM format. When read from the TPM the certificate is encoded in DER [19].

-----BEGIN CERTIFICATE-----

```
MIID7zCCAtegAwIBAgIBATANBgkqhkiG9w0BAQsFADAUMRIwEAYDVQQDDAlFeGft
cGxlQ0EwHhcNMTQwMTE1MTU0MDUwWhcNMTUwMTE1MTU0MDUwWjAAMIIBIjANBgkq
hkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAnvcvm0aOBK05rdNInYXzJGV5SFteVUFpt
XFxg4evROvlulB3BzUmFGQYFDcItVnJX2fAvf0UJLtLBVBQggb5ylL6bRpj72cS3
oyNbs0CGmix9Z1QDjkZzFvIsD1GcK00tvsCvsEItH8Cm0fq8WcGFijWldRD5eulP
55pqlbAHAvIo4+VLMJVBG71xrKGZeHPjKoq6seYjh7AGy+hk2vmFzpZ8Ghdgqv+K
02IZ7FEduylHW8U3qsxBHysMut4inj6AiVf4670Os5meHiifIK9MGkovMrFY9iX
uUVUs/KXpElsgEOX9BLvx1BPcODosr5K+z5i71OtIXy4CXrPvcGzRwIDAQABo4IB
XjCCAVowQAYIKwYBBQUHAQEENDAYMDAGCCsGAQUFBzAChiRodHRwOi8vd3d3LmV4
YW1wbGUuY29tL0V4YW1wbGVdQS5jcncQwDgYDVR0PAQH/BAQDAgAgMFkGA1UdEQEB
/wRPME2kSzBJMRyWfAYFZ4EFAGEMC2lkOjU0NDM0NzAwMRcwFQYFZ4EFAGIMDEFC
Q0RFRjEyMzQ1NjEWMWBBQWwEBAQIDDAtpZDowMDAxMDAyMzAMBGNVHRMBAf8EAjAA
MDUGA1UdHwQuMCwwKqAooCaGJGh0dHA6Ly93d3cuZXhhbXBsZS5jb20vRXhhbXBs
ZUNBLmNybdAQBgNVHSAECTAHMAUGAyoDBDAfBgNVHSMEGDAWgBQ0d2ckTESv554q
4LJMaVeVJLM92jAQBgNVHSECTAHBgVngQUIATAhBgNVHQkEGjAYMBYGBWwEBAQIQ
MQ0wCwwDMi4wAgEAAgFjMA0GCSqGSIb3DQEBCwUAA4IBAQAba2btJ/+4z02MWpNp
99AFGPu3yIaJqI6NeHvC6fxe/9lWlHKISR+CnpAh03/MKT8TP2/cUSi0jkkQNh
MtrueUNofE79fYXtHXHU7wzzUFWNwCmhTuHDYl3jmd0fJ9yA2CuUHT6q3UV+PwXN+
EHElhQwC8QtNC/5A7wY1e5dBLdgwSSIgTc4lSsbNcZ9d+m7mWEWpumSYU0czTDEN
HmdU/VJuDN/RCOAyBb+hcl9LAucGmnFYOhxWHfd9zbXZA1ldFUxrpPuVfKx+Eo8f
rMsB2oZKMwSYUAWotqolhLe2wdBMRjdmVz44kIhuFB7y4BpQjlB1+xAzX9Hb31CG
eoS2
```

-----END CERTIFICATE-----

B.2 Example 2 (firmware-limited EK)

This annex provides an example for the ASN.1 encoding of a standard tcg-dice-TcbInfo extension present in a firmware-limited TPM Endorsement Key Certificate. The values used in this example are for illustrative purposes and must be replaced with manufacturer-specific data.

```
// SET
31 5C
  // SEQUENCE
  30 5A
    // OBJECT IDENTIFIER tcg-dice-TcbInfo (2.23.133.5.4.1)
    06 06 67 81 05 05 04 01
    // BOOL (critical = 0)
    01 01 00
    // OCTET STRING
    04 50
      // SEQUENCE
      30 4E
        // CONTEXT_SPECIFIC | version (02)
        // version = ABCDEF123456
        82 0C 41 42 43 44 45 46 31 32 33 34 35 36
        // CONTEXT_SPECIFIC | svn (03)
        // svn = 4
        83 01 04
        // CONTEXT_SPECIFIC | CONSTRUCTED | fwids (06)
        A6 2F
          // SEQUENCE
          30 2D
            // OBJECT IDENTIFIER SHA256
            06 09 60 86 48 01 65 03 04 02 01
            // OCTET STRING
            // fwid = 0x0000...00
            04 20 00 00 00 00 00 00 00 00 00 00
              00 00 00 00 00 00 00 00
              00 00 00 00 00 00 00 00
              00 00 00 00 00 00 00 00
            // CONTEXT_SPECIFIC | vendorInfo (08)
            // vendor info = 0x00000000
            88 04 00 00 00 00
            // CONTEXT_SPECIFIC | type (09)
            // type = 0x00000000
            89 04 00 00 00 00
```